



PHILADELPHIA
— SEPTEMBER 27-30, 2026 —
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Skeletal muscle alterations in children with CP

Sudarshan Dayanidhi, PT, PhD
Sept 27th 2026

Shirley Ryan
Abilitylab®

Northwestern
University

FINANCIAL DISCLOSURE
AACPDM 80th Annual Meeting
September 27-30, 2026



Speaker Name: Sudarshan Dayanidhi

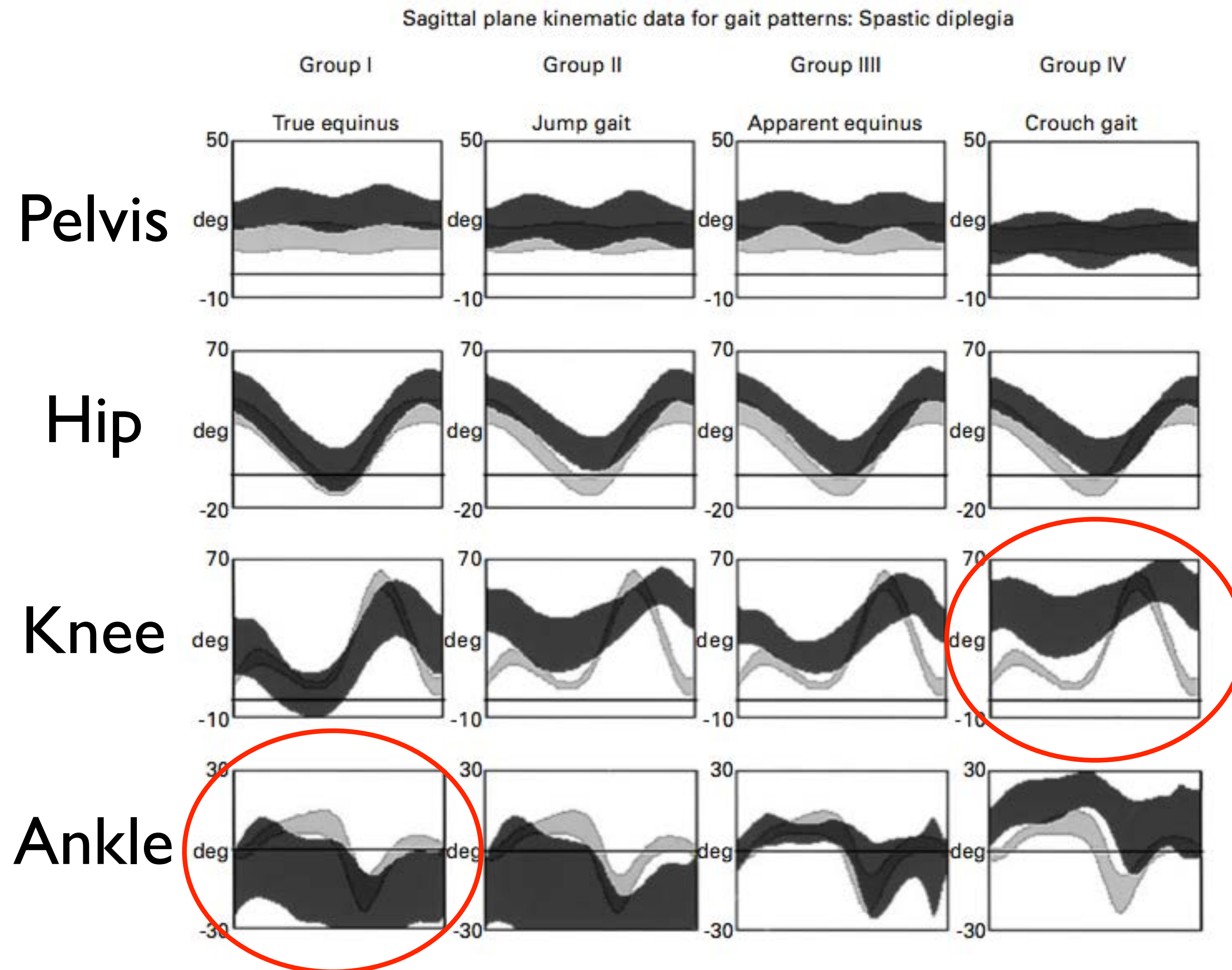
1. Disclosure of Relevant Financial Relationships

I have no financial relationships to disclose.

2. Disclosure of Off-Label and/or investigative uses:

I will not discuss off label use and/or investigational use in my presentation.

Gait and Contractures



Muscle biopsy based studies in children with CP

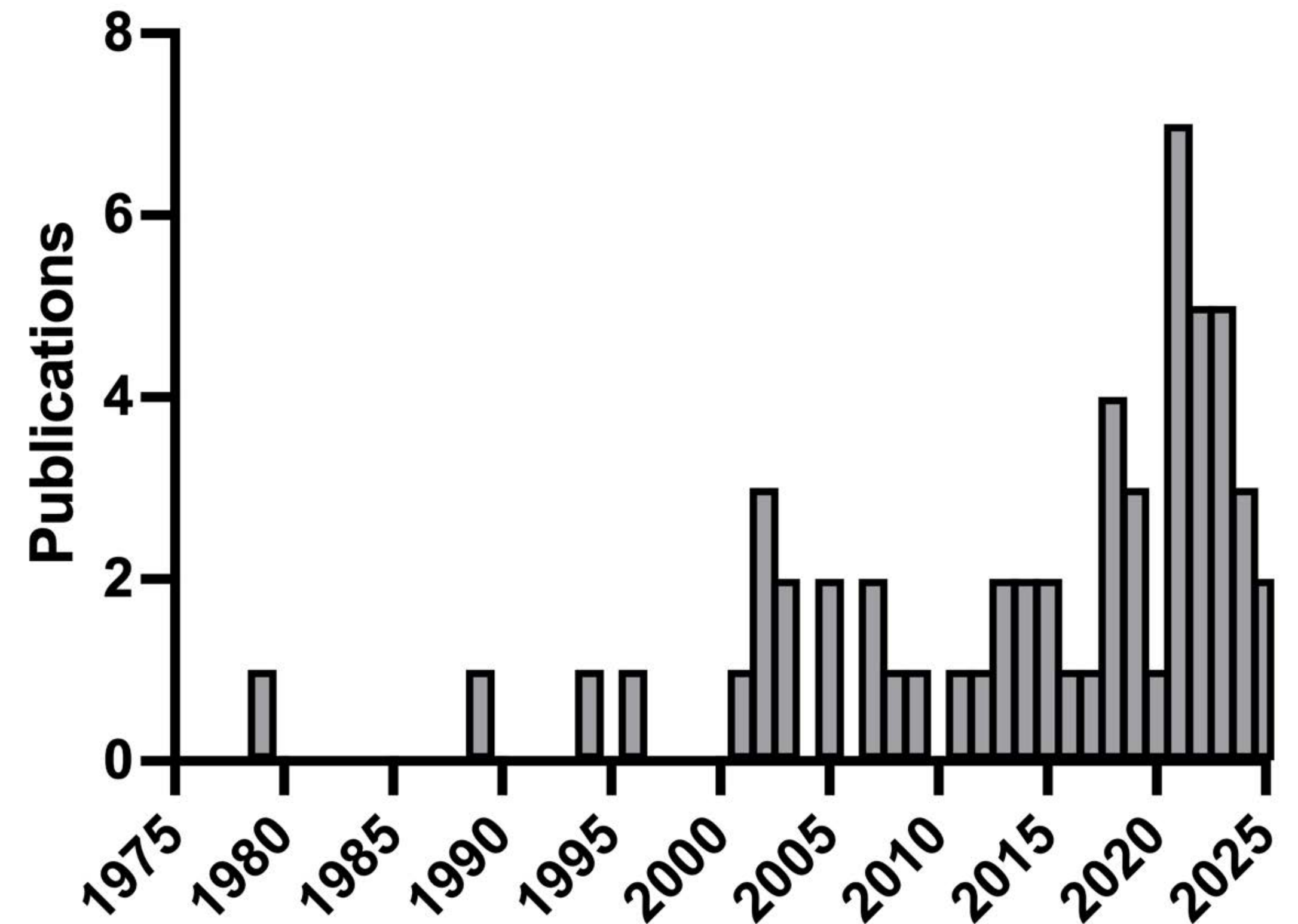
Pathology of Spastic Muscle in Cerebral Palsy

M. E. CASTLE, M.D.,* THEODORE A. REYMAN, M.D.,**
AND MARIE SCHNEIDER, B.S.†

Received: September 1, 1978.

Clin Orthop Relat Res. 1979(142):223-32.

- n=85 (108 biopsies across various muscles)





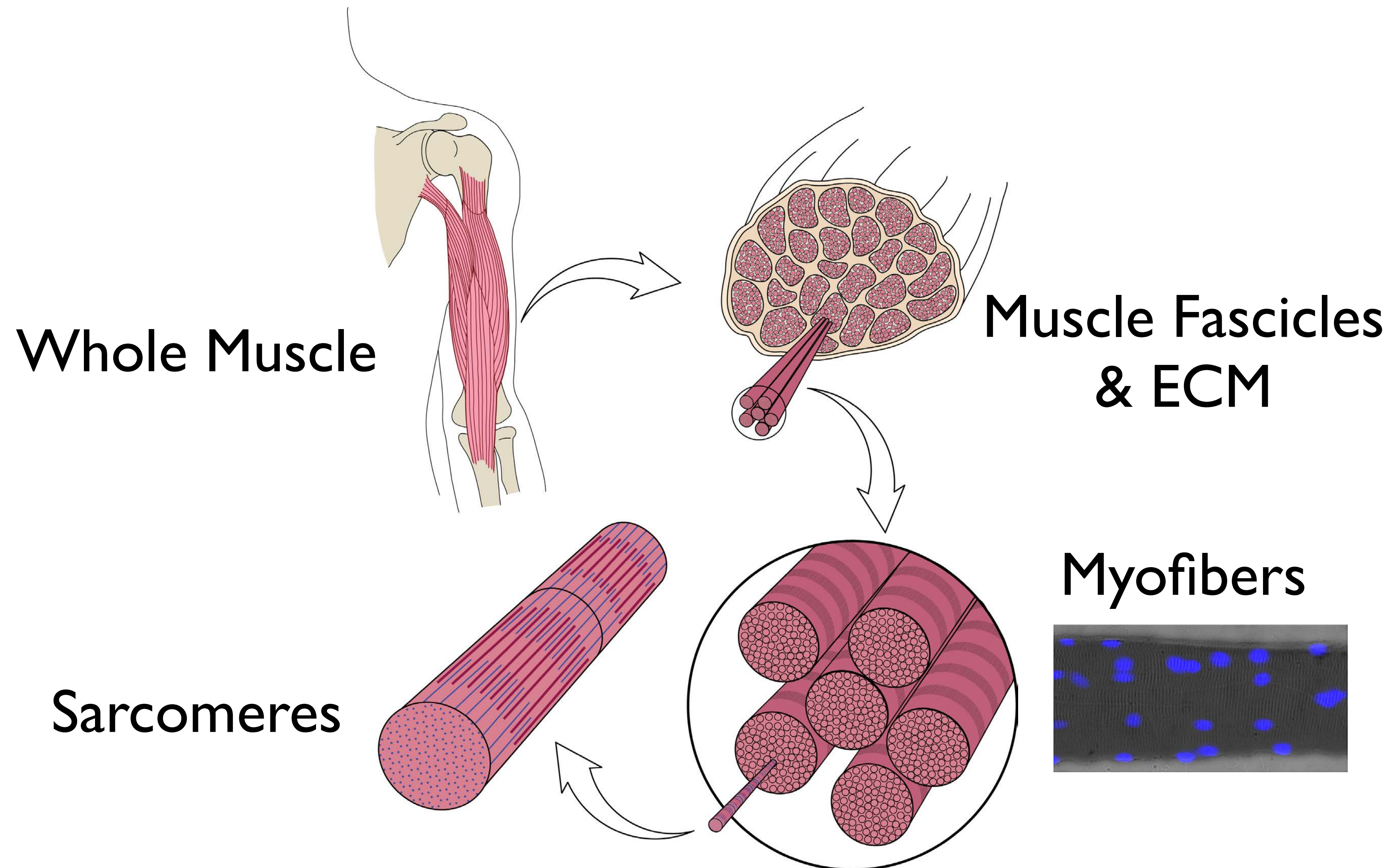
Microscopic and molecular aspects of skeletal muscle alterations in cerebral palsy

Sebastian Edman^{1,2,3}  | Oscar Horwath^{1,3}  | Eva Pontén^{1,4}  |
Sudarshan Dayanidhi^{5,6}  | Ferdinand von Walden¹ 

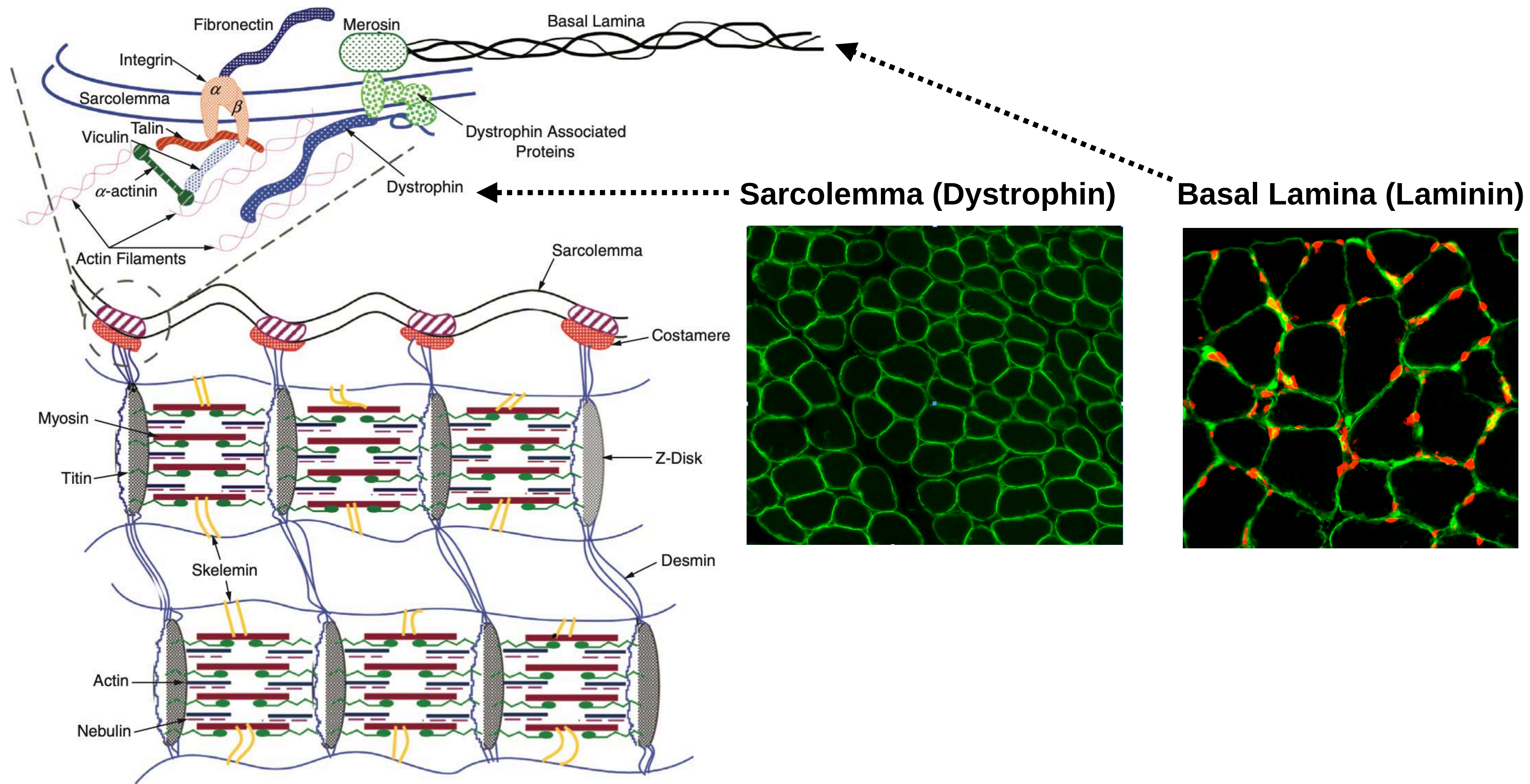
Dev Med Child Neurol. 2026;68:343–357.

- Muscle changes in children with CP are complex and multifactorial!!

Muscle Structure

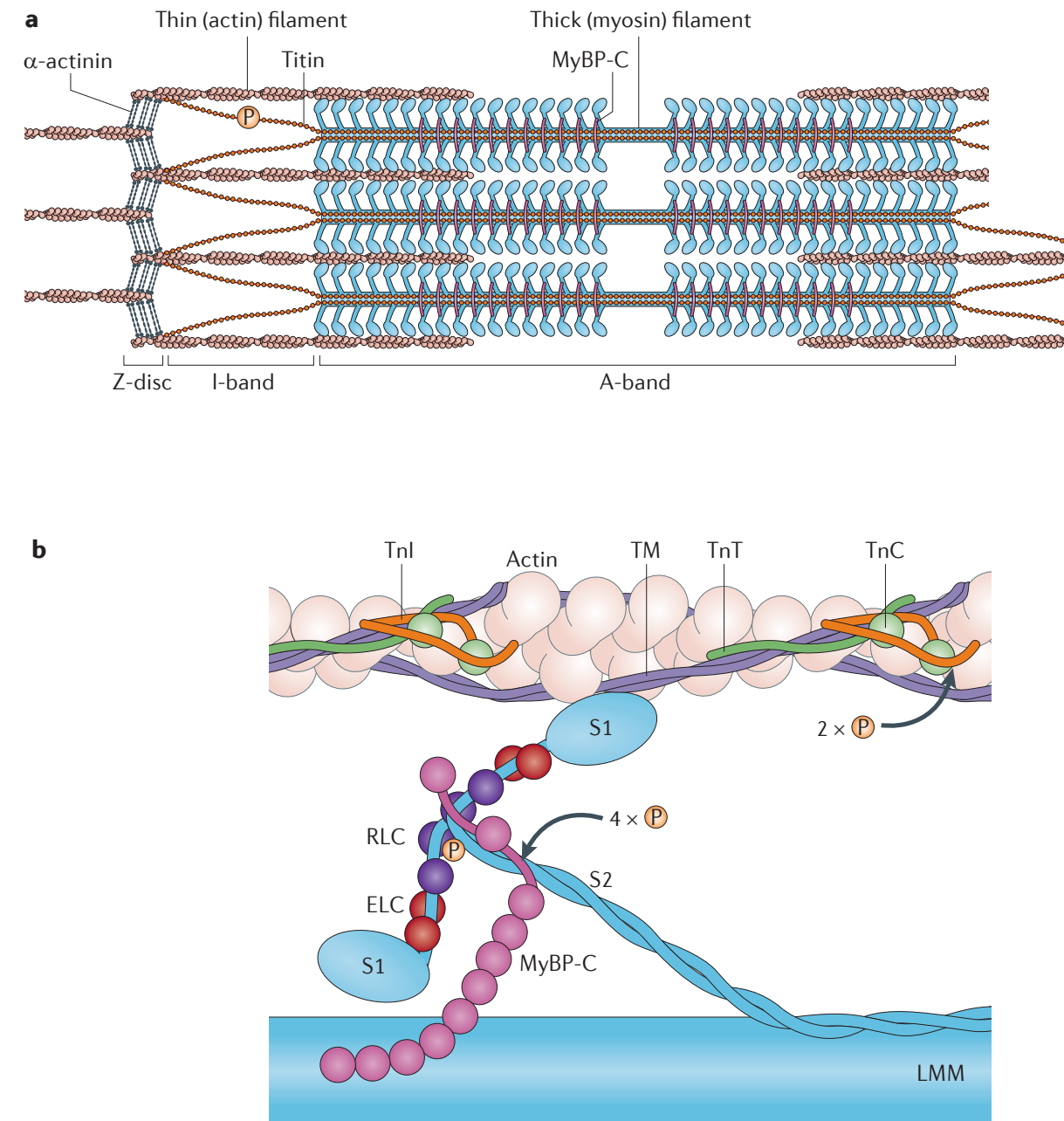


ECM -Structural Proteins

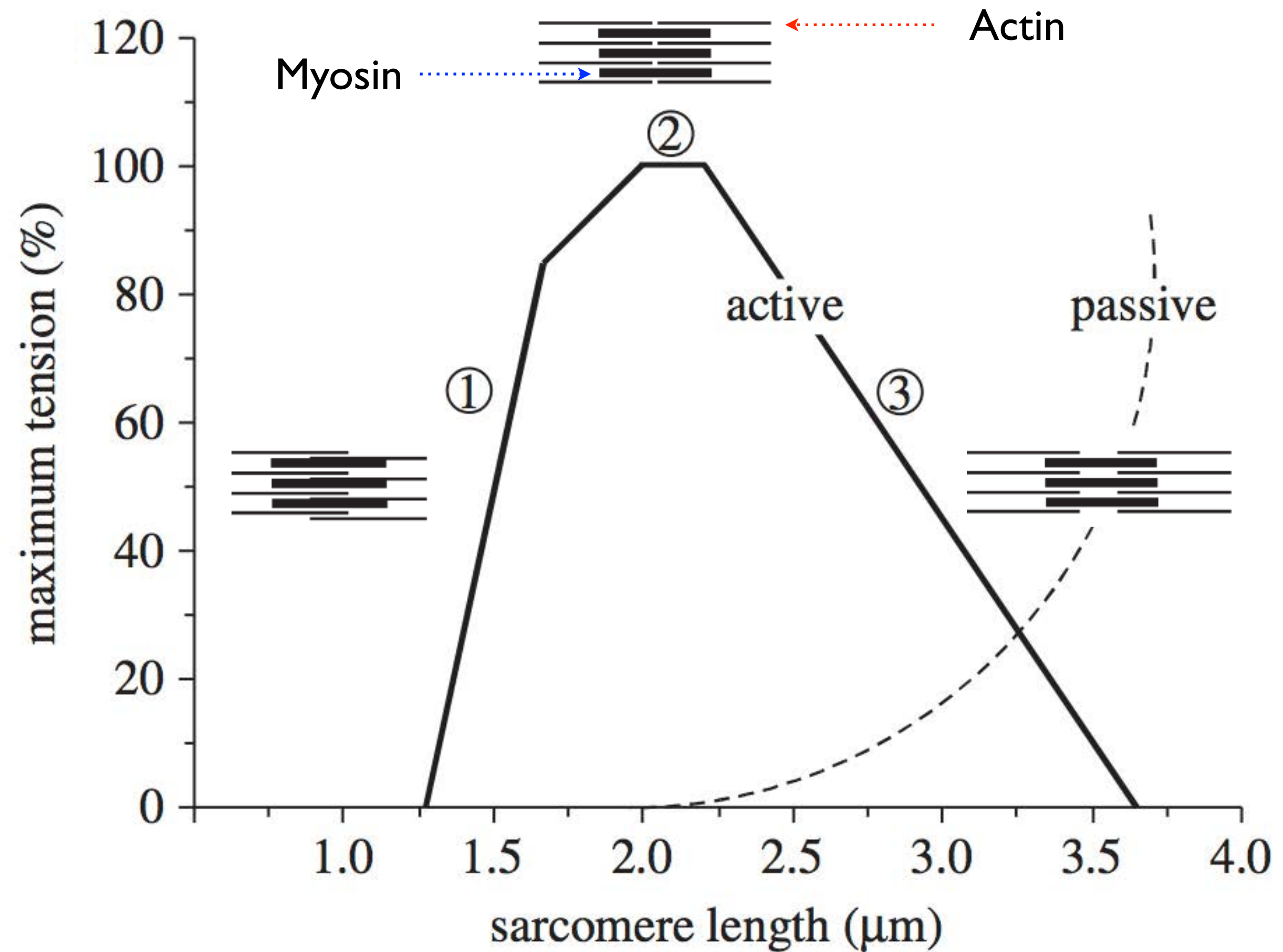


- Numerous proteins connect the myofibrils to the ECM

Sarcomeres (Force-Length)



Hwang and Sykes 2015



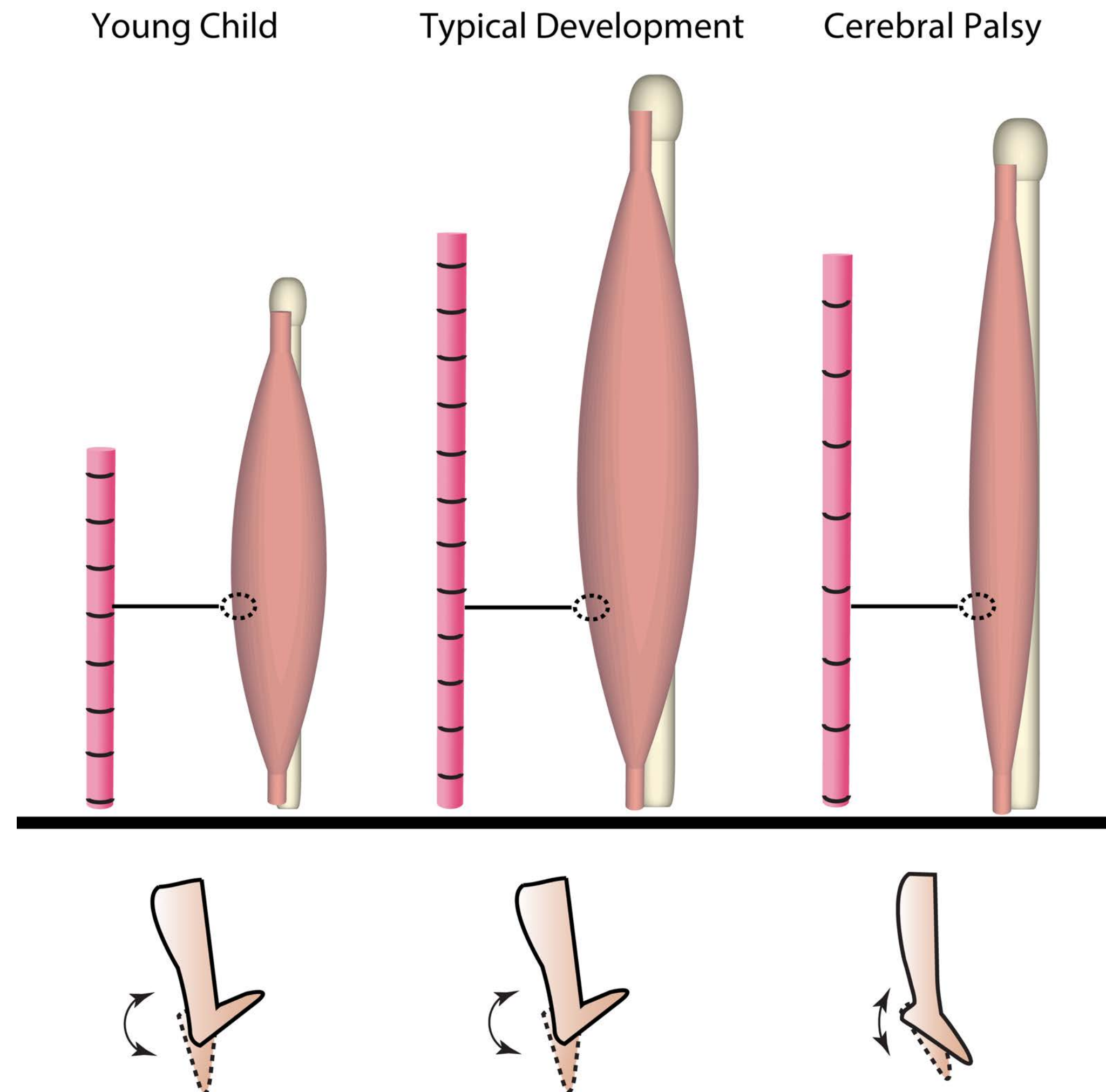
Sarcomere length dictates force production

Lieber and Ward 2011

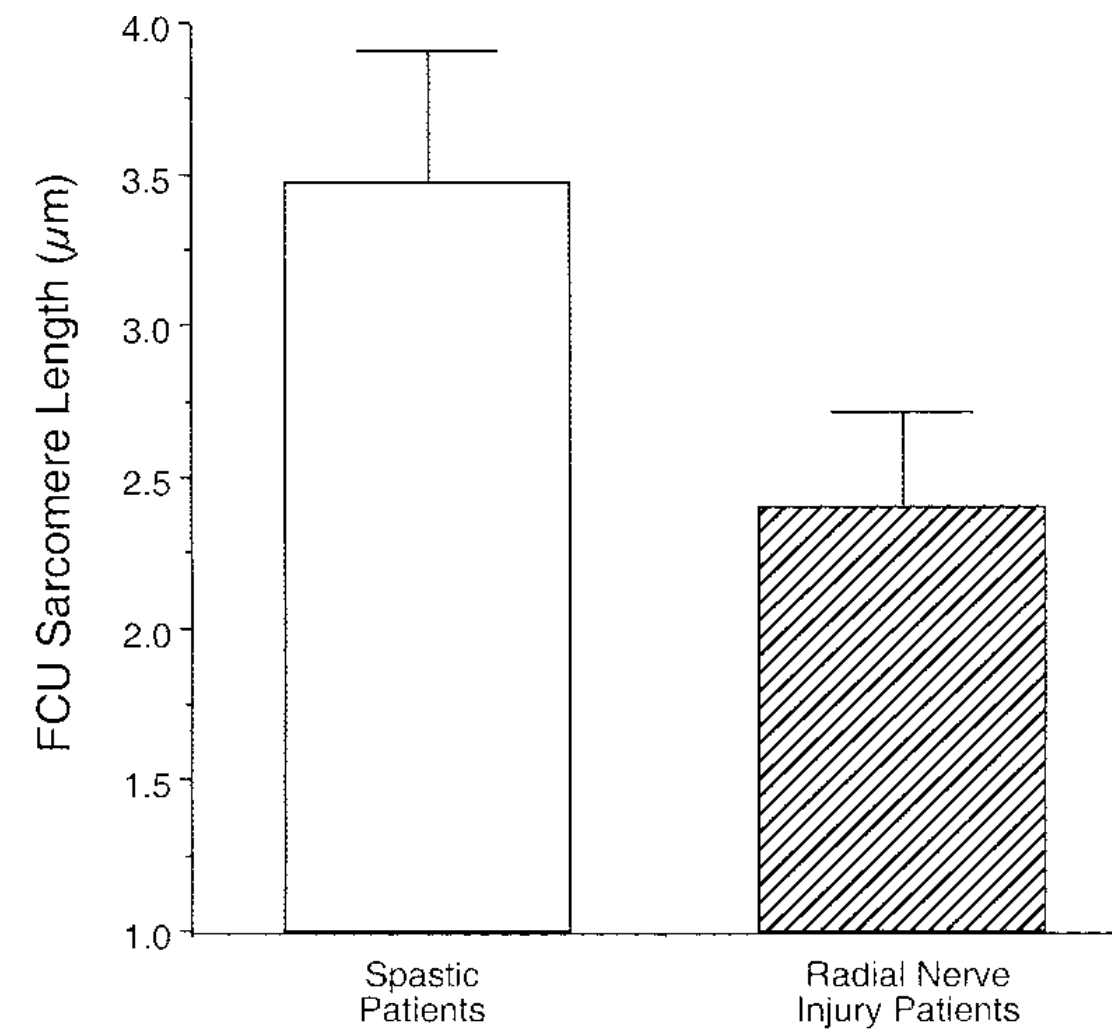
Postnatal Development

Longitudinal
muscle growth
by adding
sarcomeres
(in series)

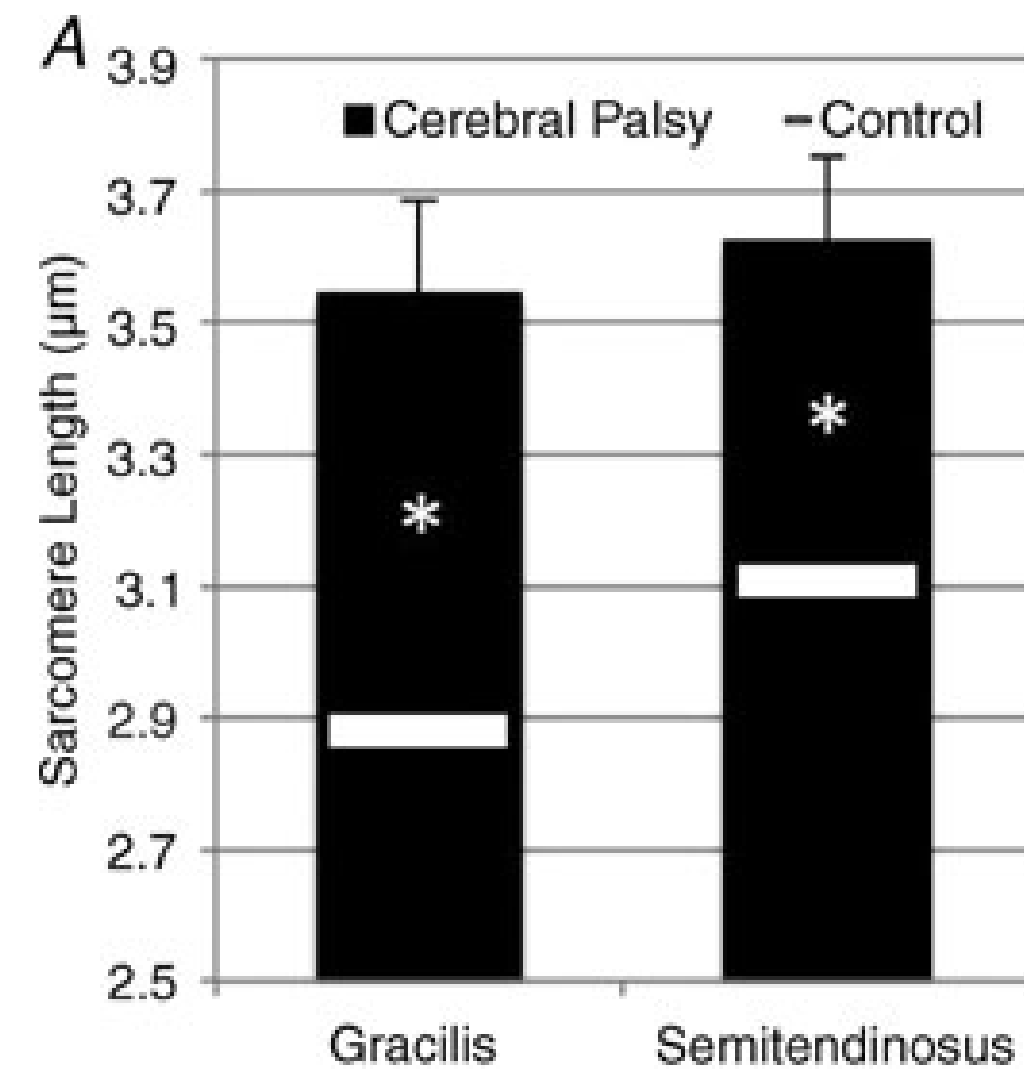
Maintains ROM



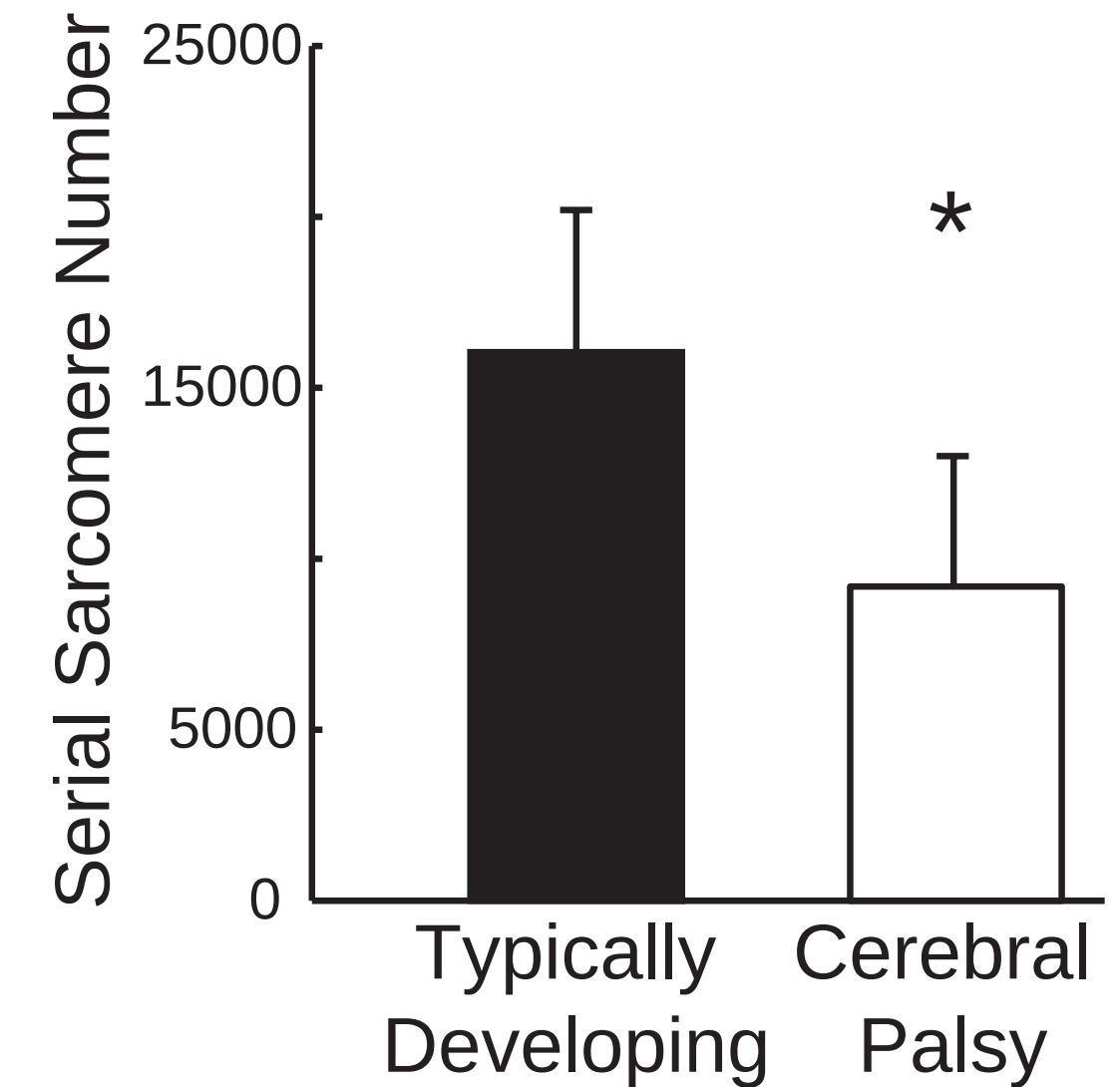
Contractures-Sarcomeres



Lieber & Fridén 2002



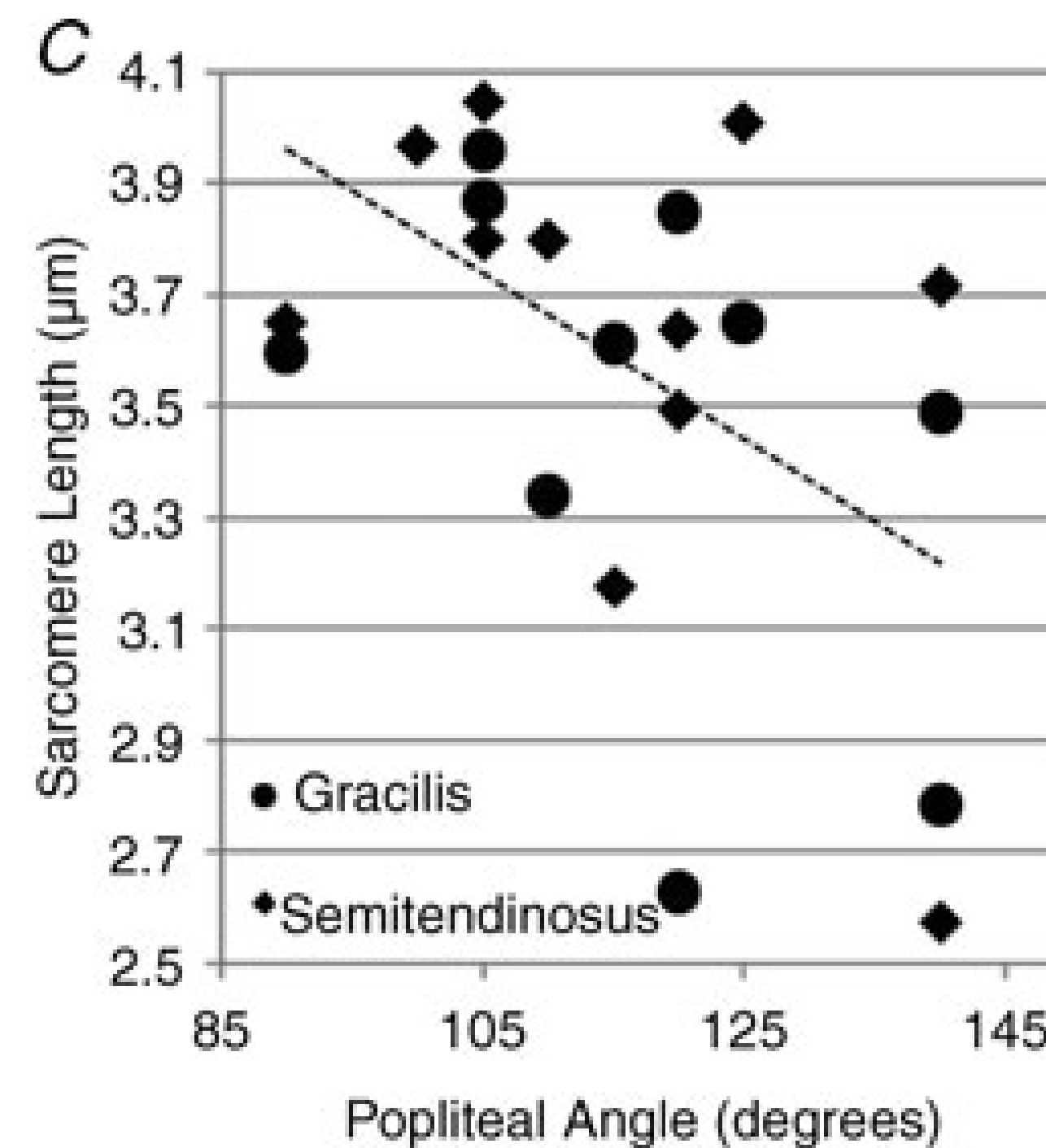
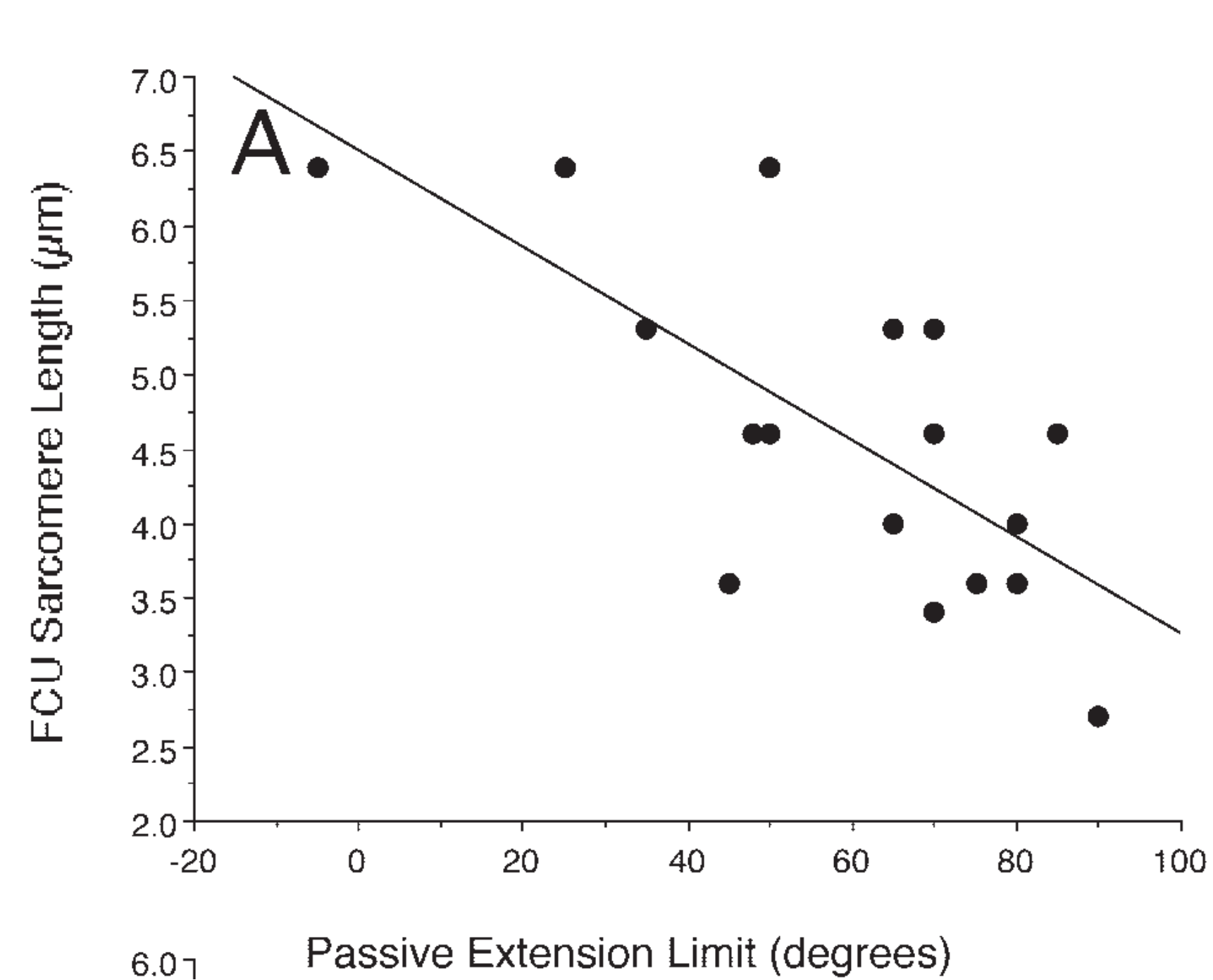
Smith et al 2011



Mathewson et al 2014

- Reduced sarcomere number
- Increased sarcomere length

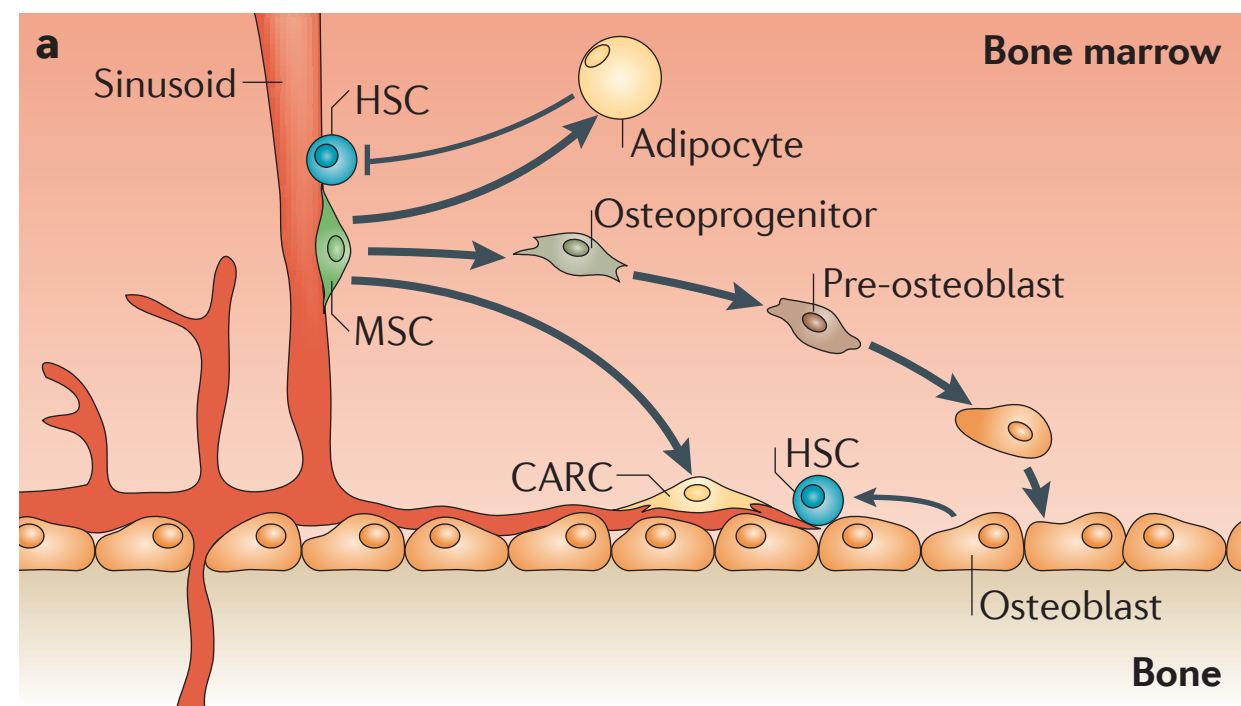
Contractures- Sarcomere Length



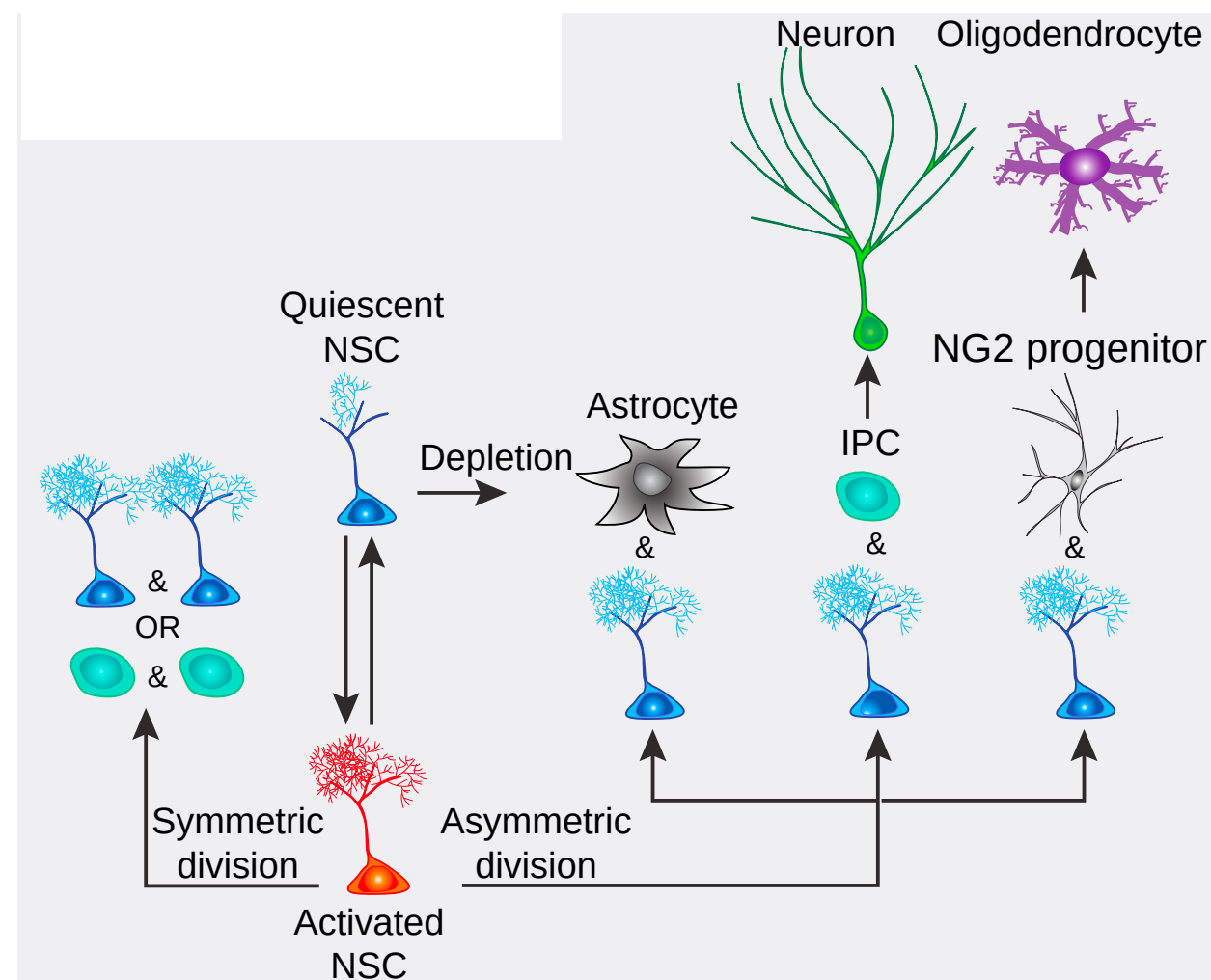
Contractures/Maximal ROM associated with sarcomere length

Stem Cells

Mesenchymal Stem Cells

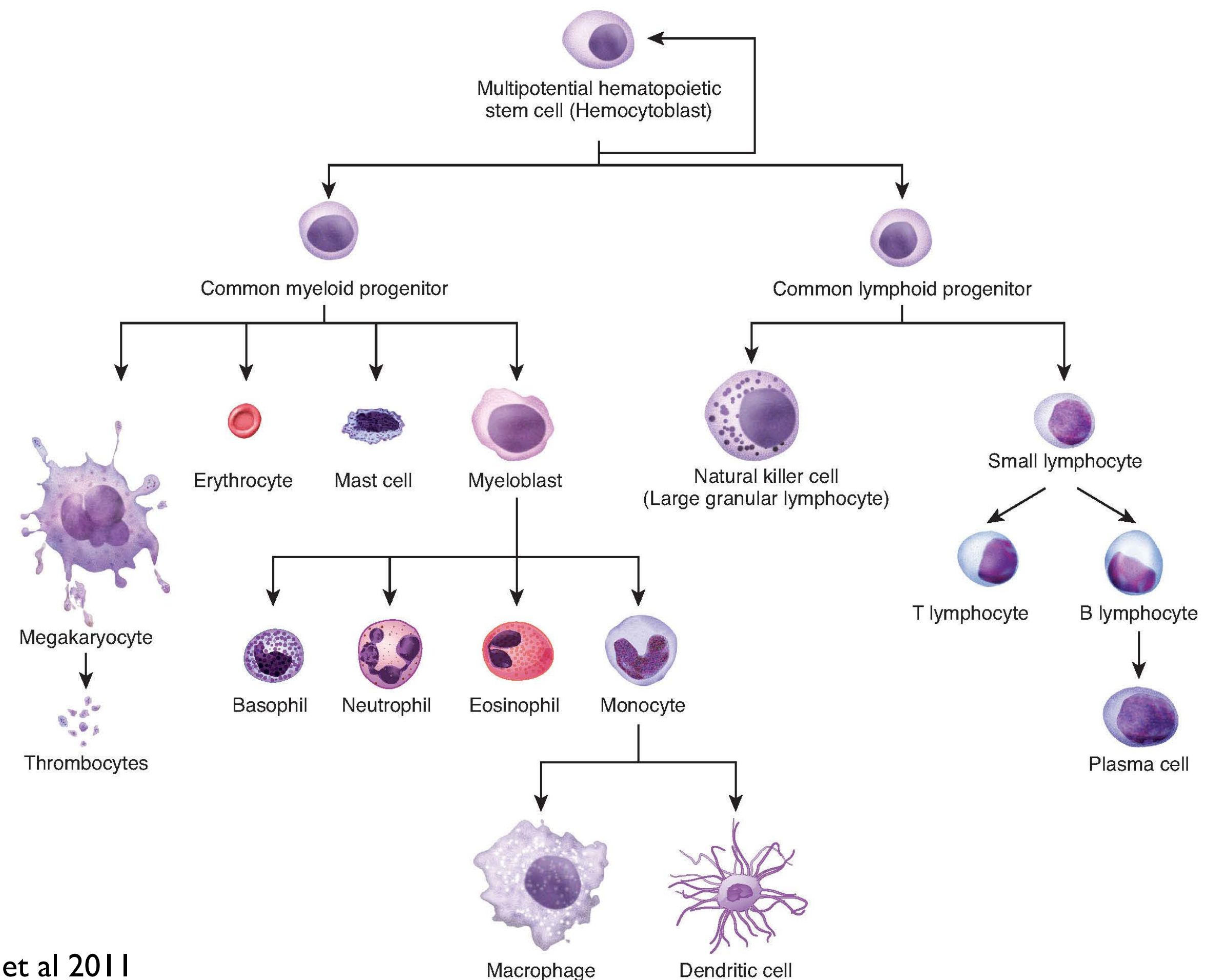


Neural Stem Cells



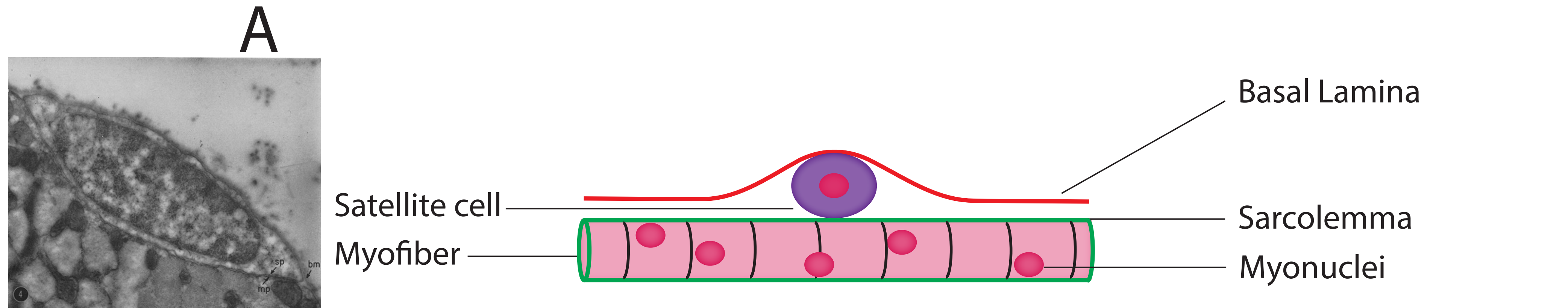
Bond et al 2015

Hematopoietic Stem Cells

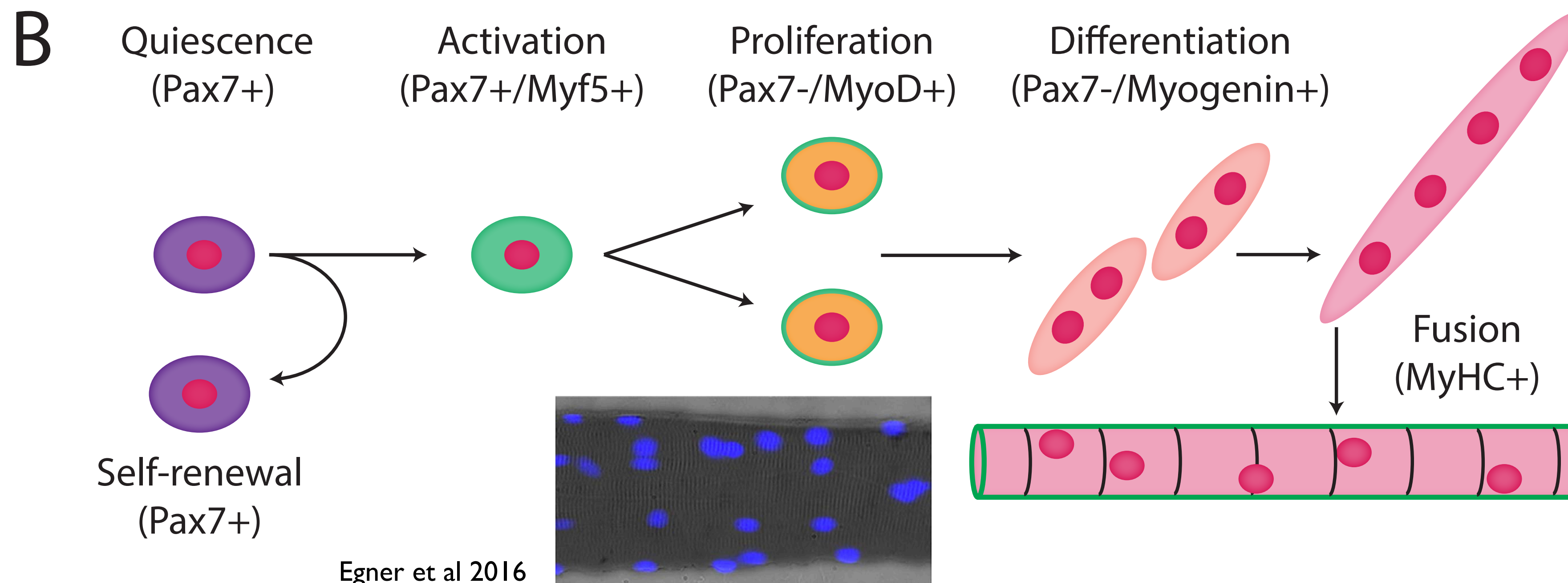


Nombela-Arrieta et al 2011

Satellite Cells=Muscle Stem Cells (MuSC)

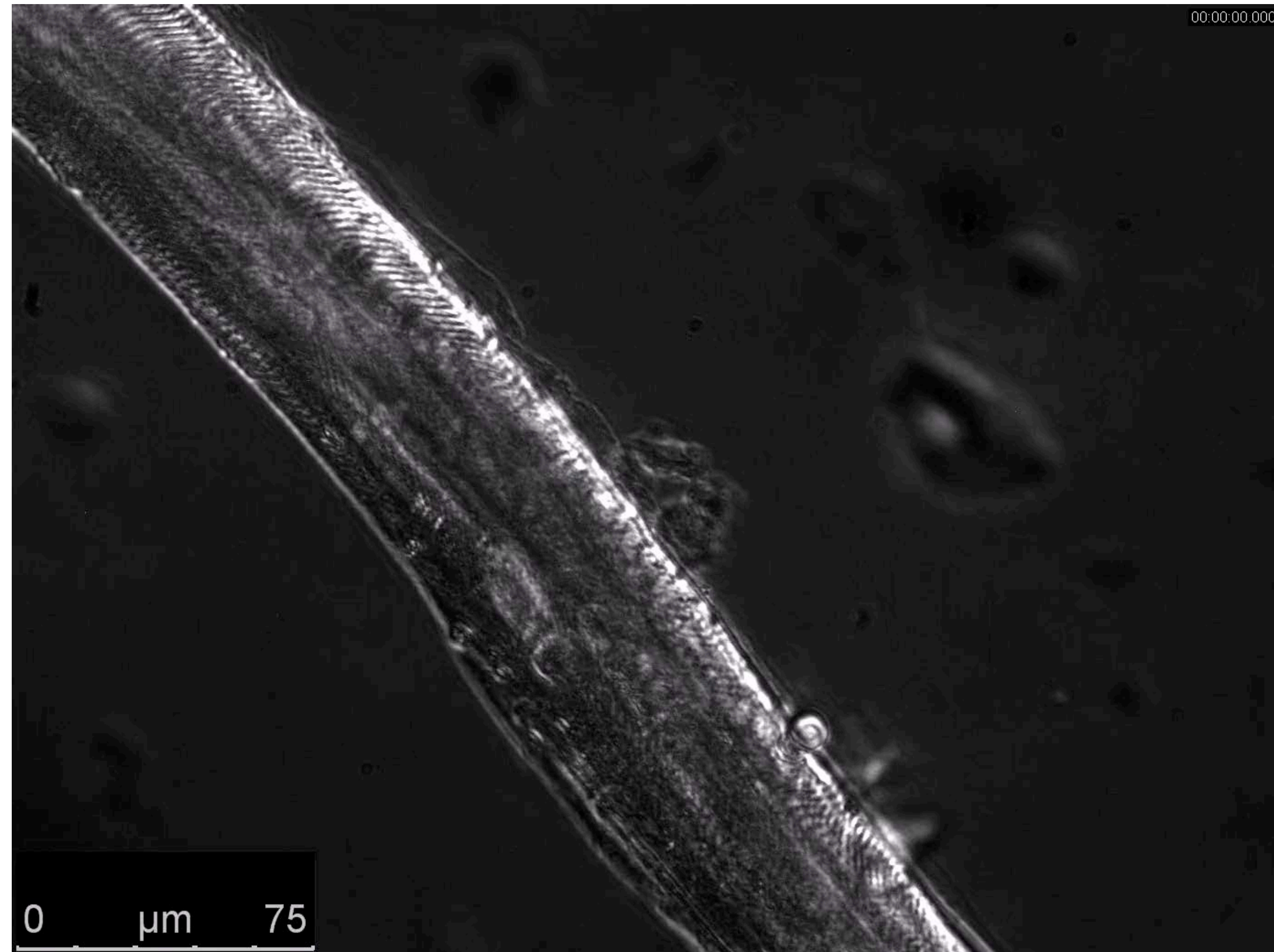


Mauro 1961



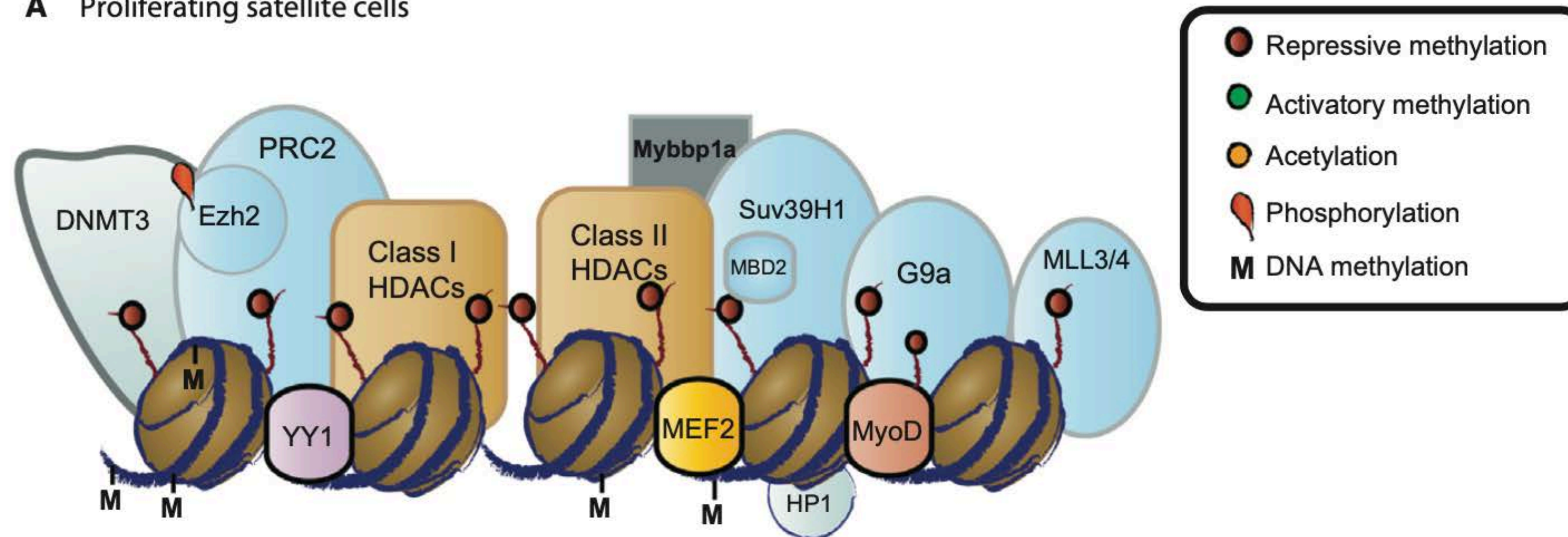
Dayanidhi and Lieber 2014

Activated SC on a fiber ex vivo



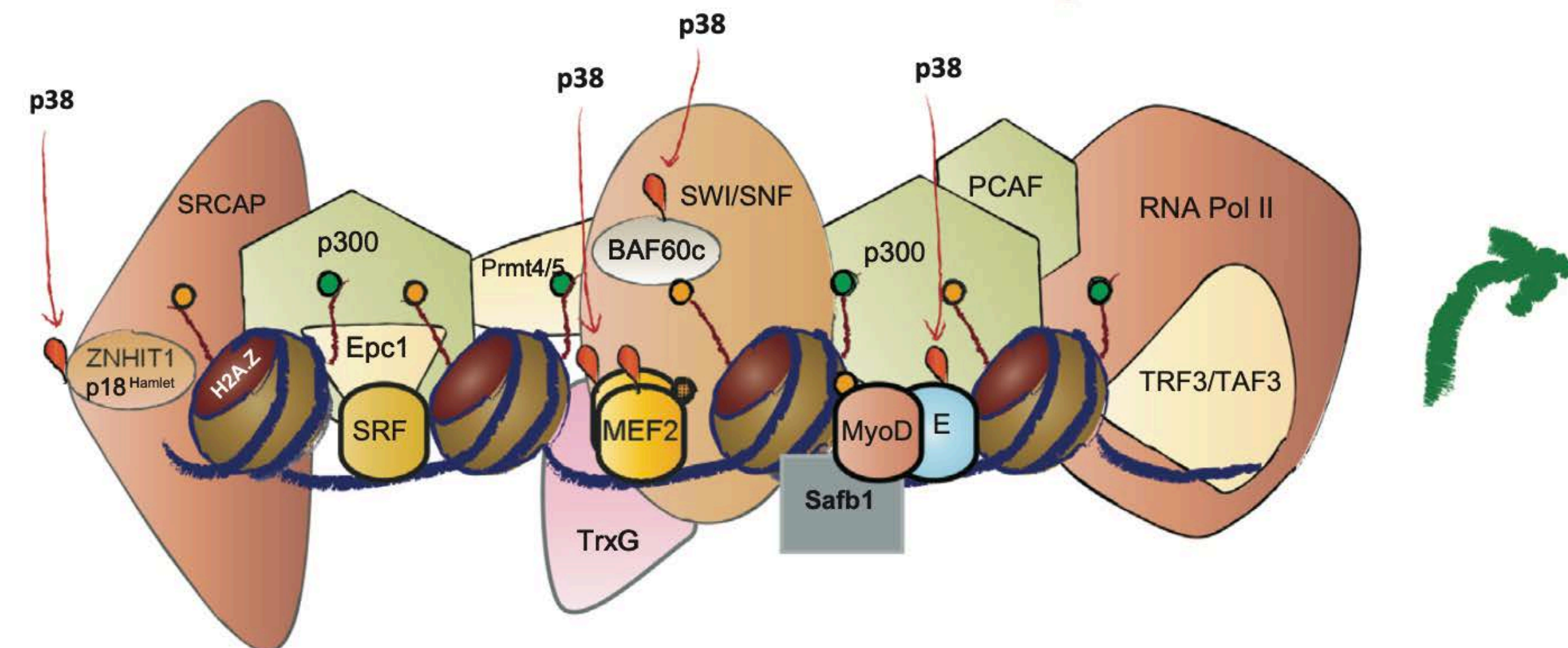
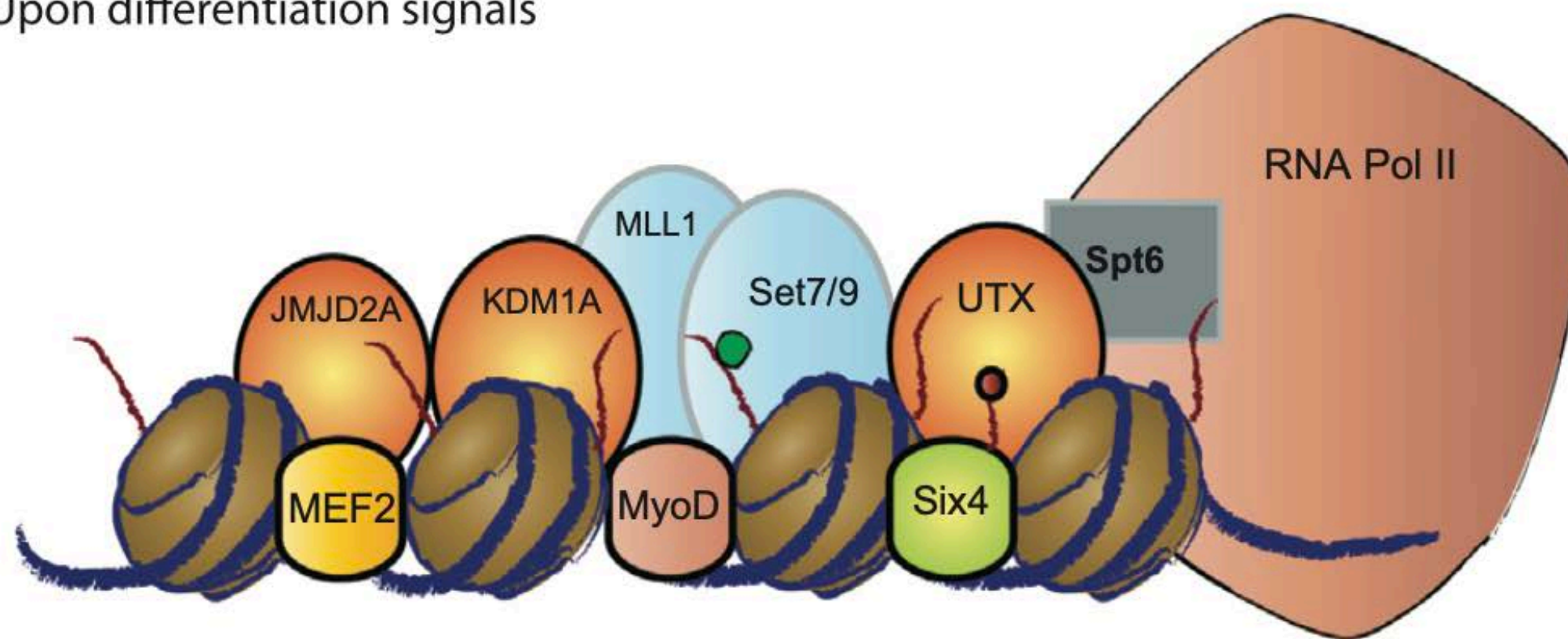
Epigenetic Processes

A Proliferating satellite cells



Specific timely complex epigenetic processes have to occur during quiescence -> activation -> proliferation -> differentiation -> fusion

B Upon differentiation signals



RESEARCH

Open Access



Differential requirement for satellite cells during overload-induced muscle hypertrophy in growing versus mature mice

Kevin A. Murach^{1,3†}, Sarah H. White^{1,3,4†}, Yuan Wen^{2,3}, Angel Ho^{1,3}, Esther E. Dupont-Versteegden^{1,3}, John J. McCarthy^{2,3} and Charlotte A. Peterson^{1,3*} 

MuSC critical in young muscles for muscle growth

© 2018. Published by The Company of Biologists Ltd | Development (2018) 145, dev167197. doi:10.1242/dev.167197



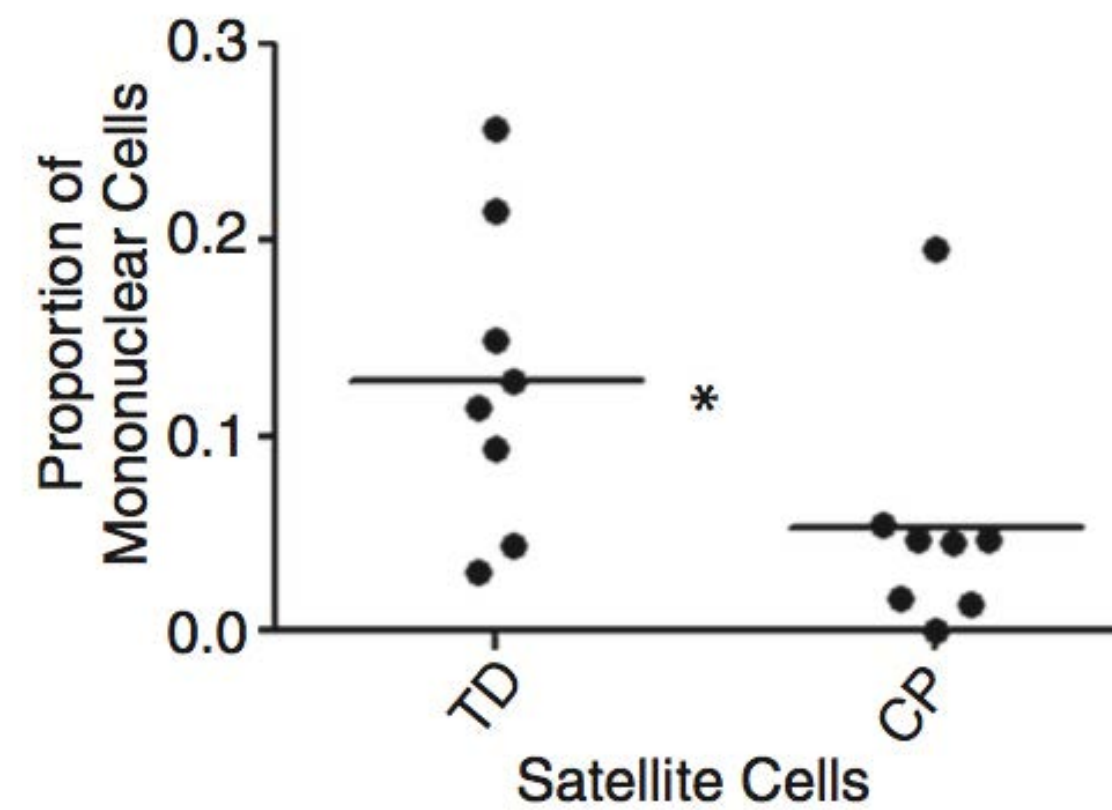
STEM CELLS AND REGENERATION

RESEARCH ARTICLE

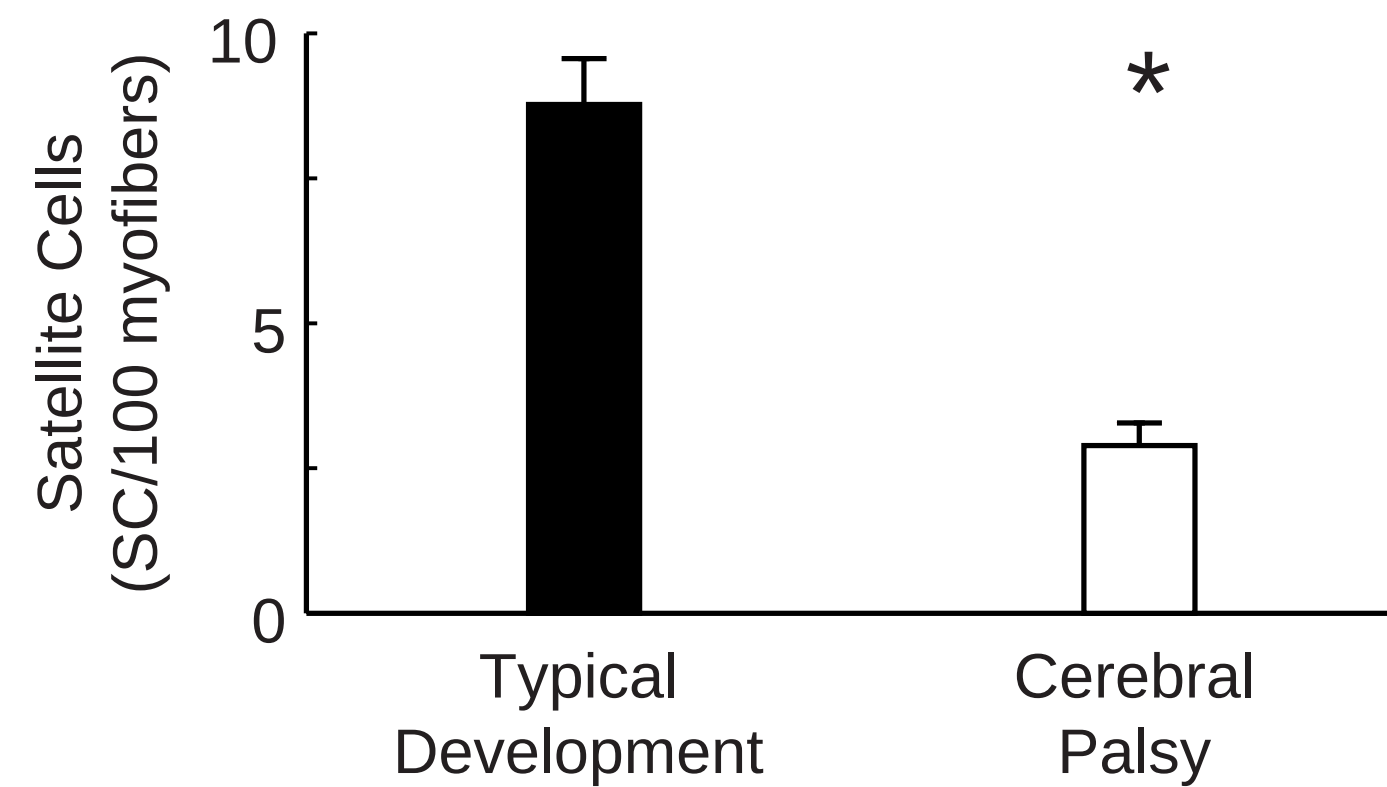
Prepubertal skeletal muscle growth requires Pax7-expressing satellite cell-derived myonuclear contribution

John F. Bachman^{1,2}, Alanna Klose², Wenxuan Liu^{2,3,*}, Nicole D. Paris², Roméo S. Blanc², Melissa Schmalz², Emma Knapp⁴ and Joe V. Chakkalakal^{2,5,‡}

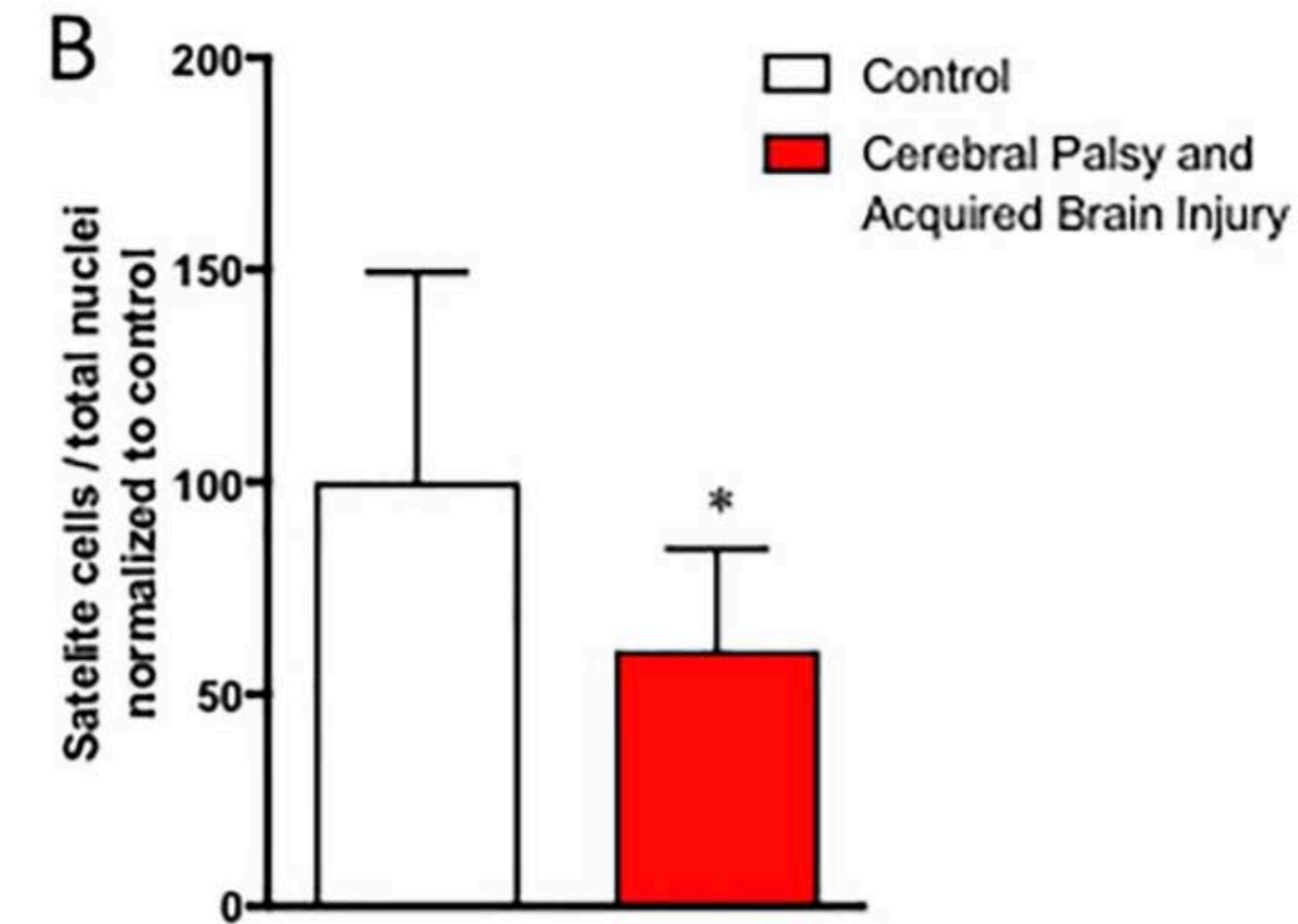
MuSC abundance compared to TD



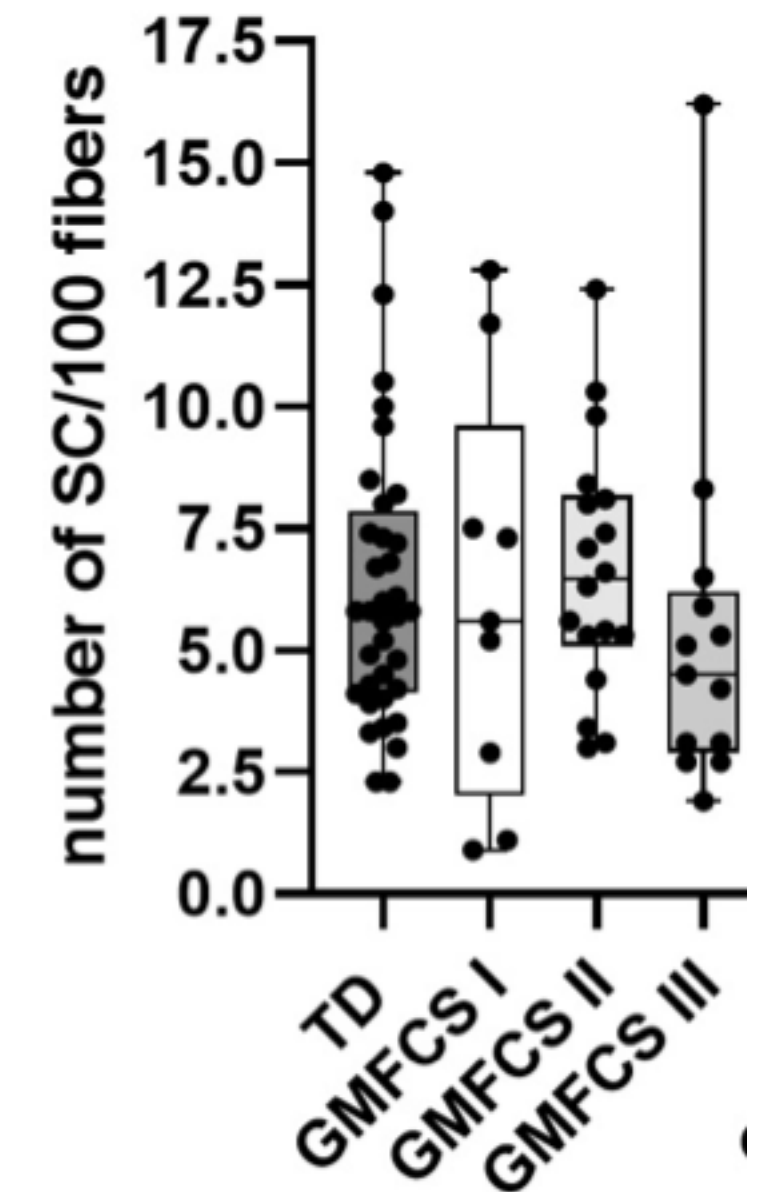
Smith et al 2013



Dayanidhi et al 2015



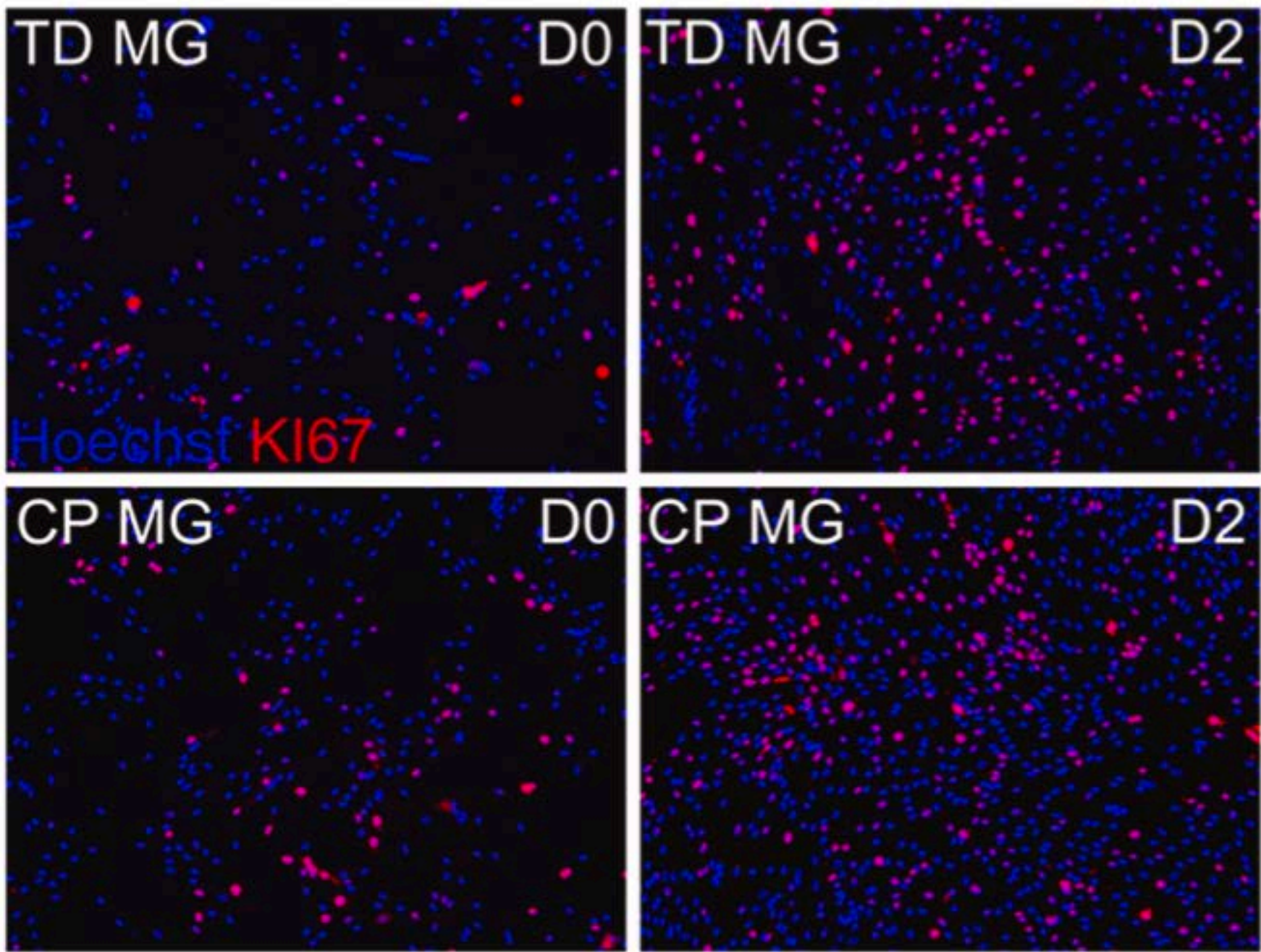
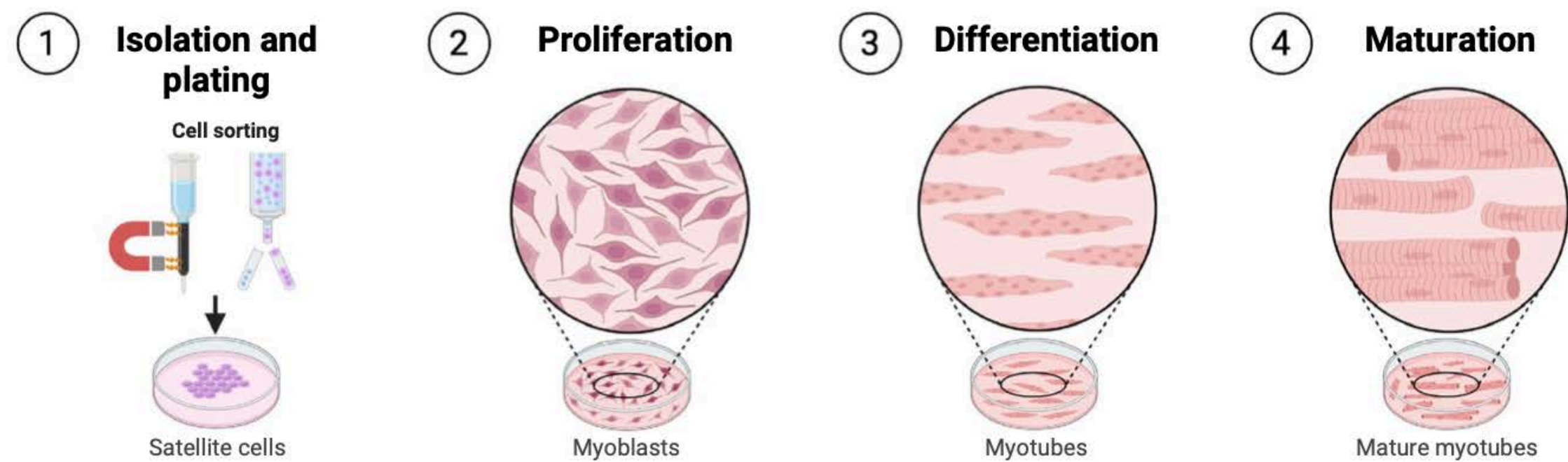
Von Walden et al 2018



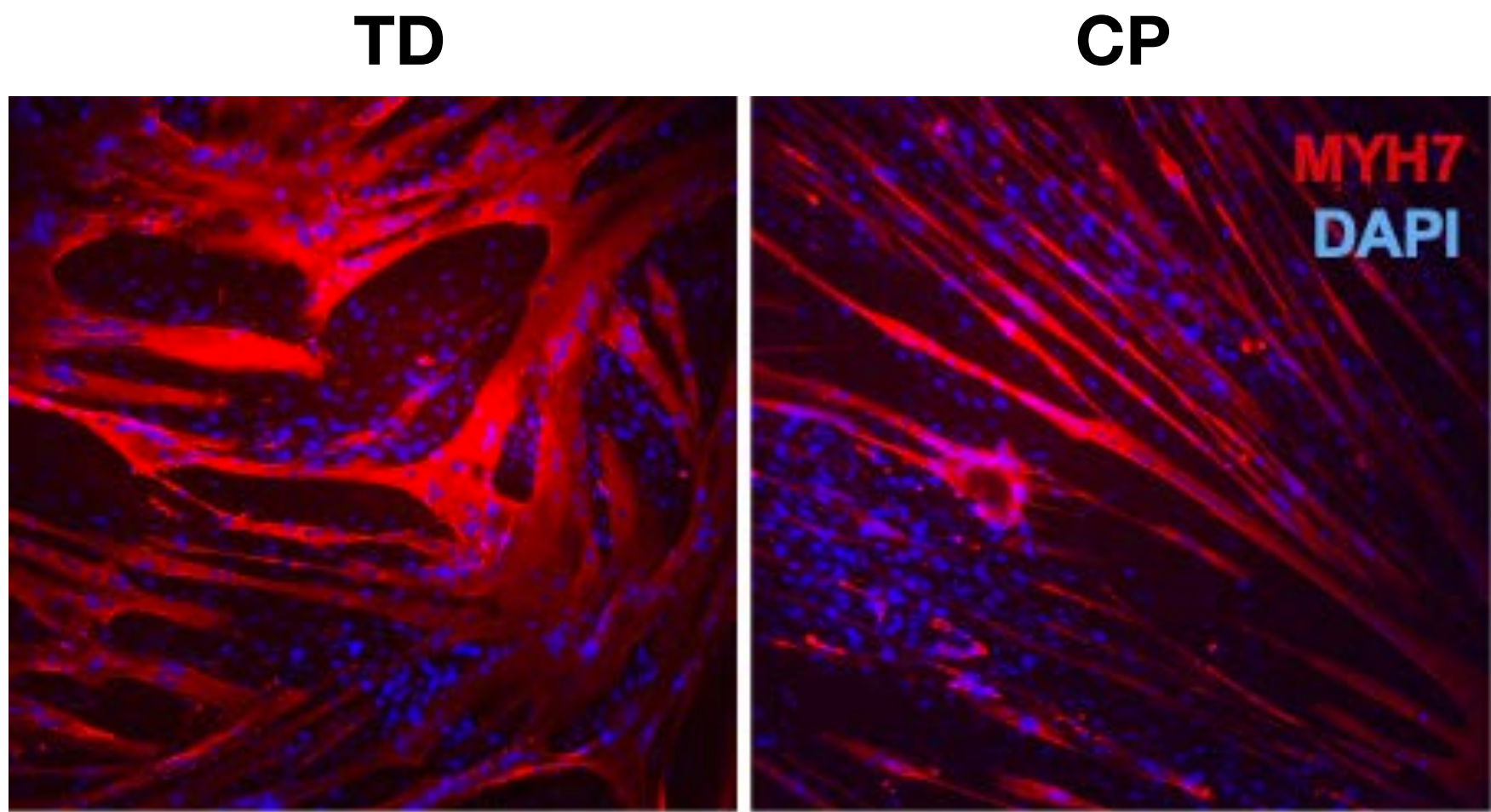
Deschrevel et al 2024

SC number reduced by 50-70% in hamstrings and biceps contractures compared to TD

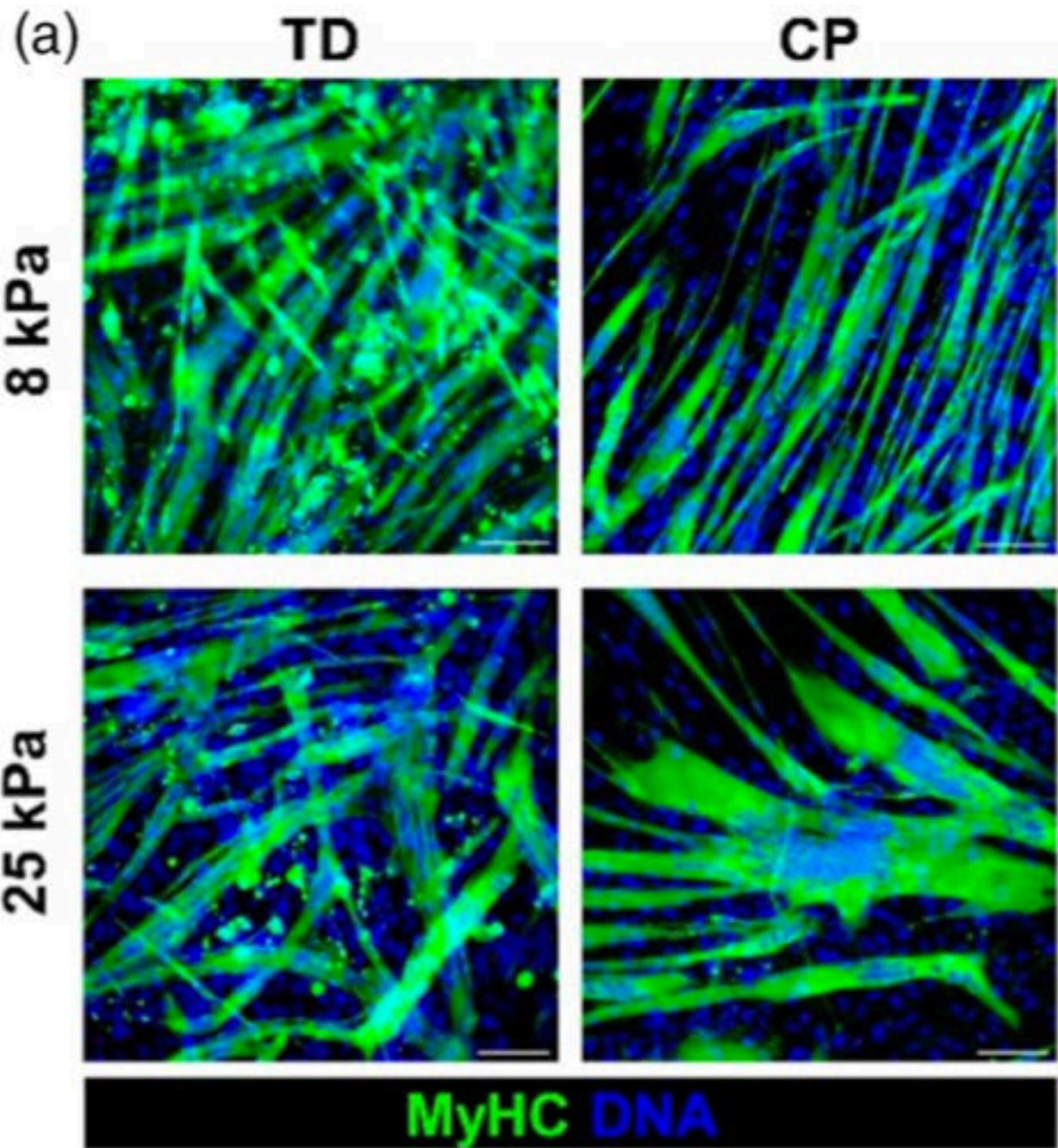
Myoblast capacity in cell culture is altered



Corvelyn et al 2023



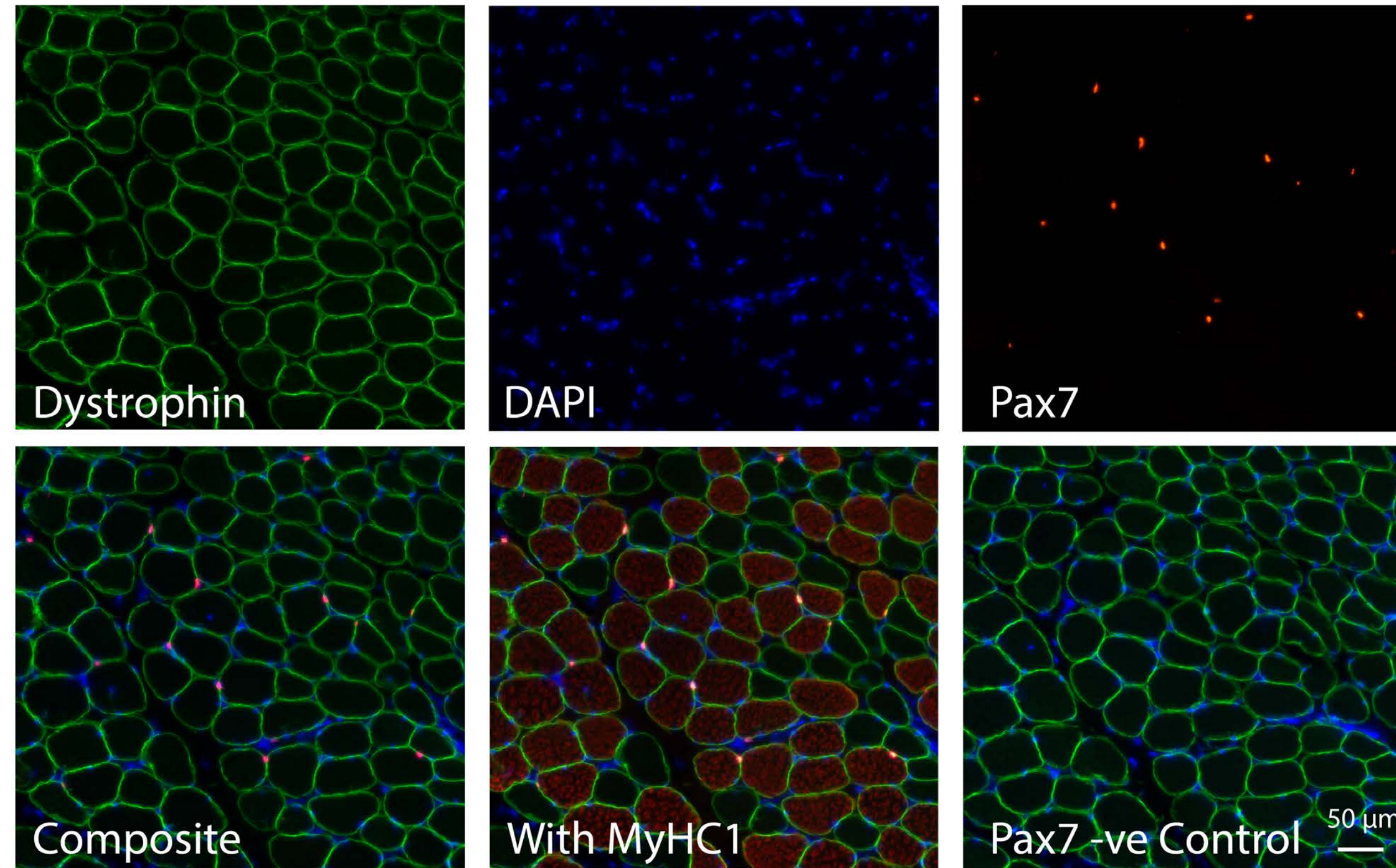
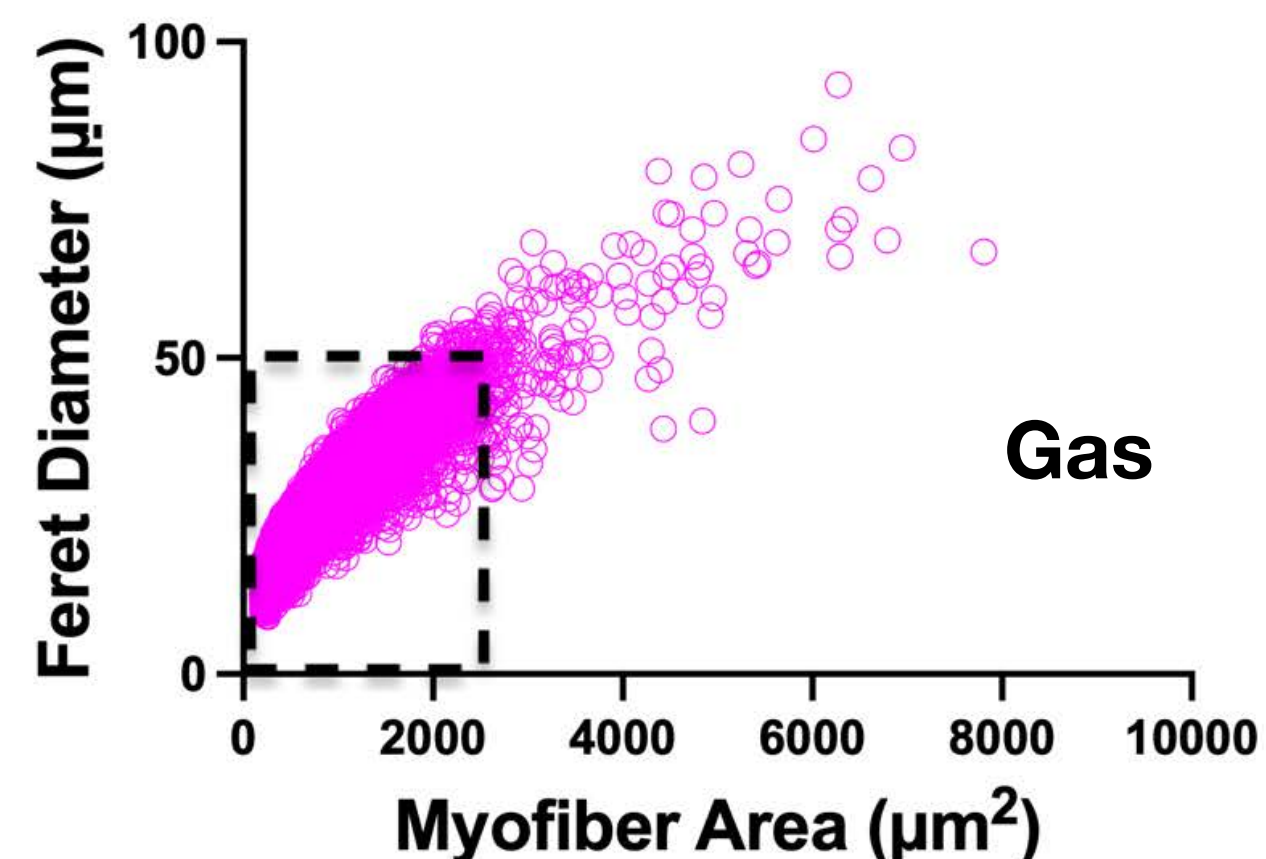
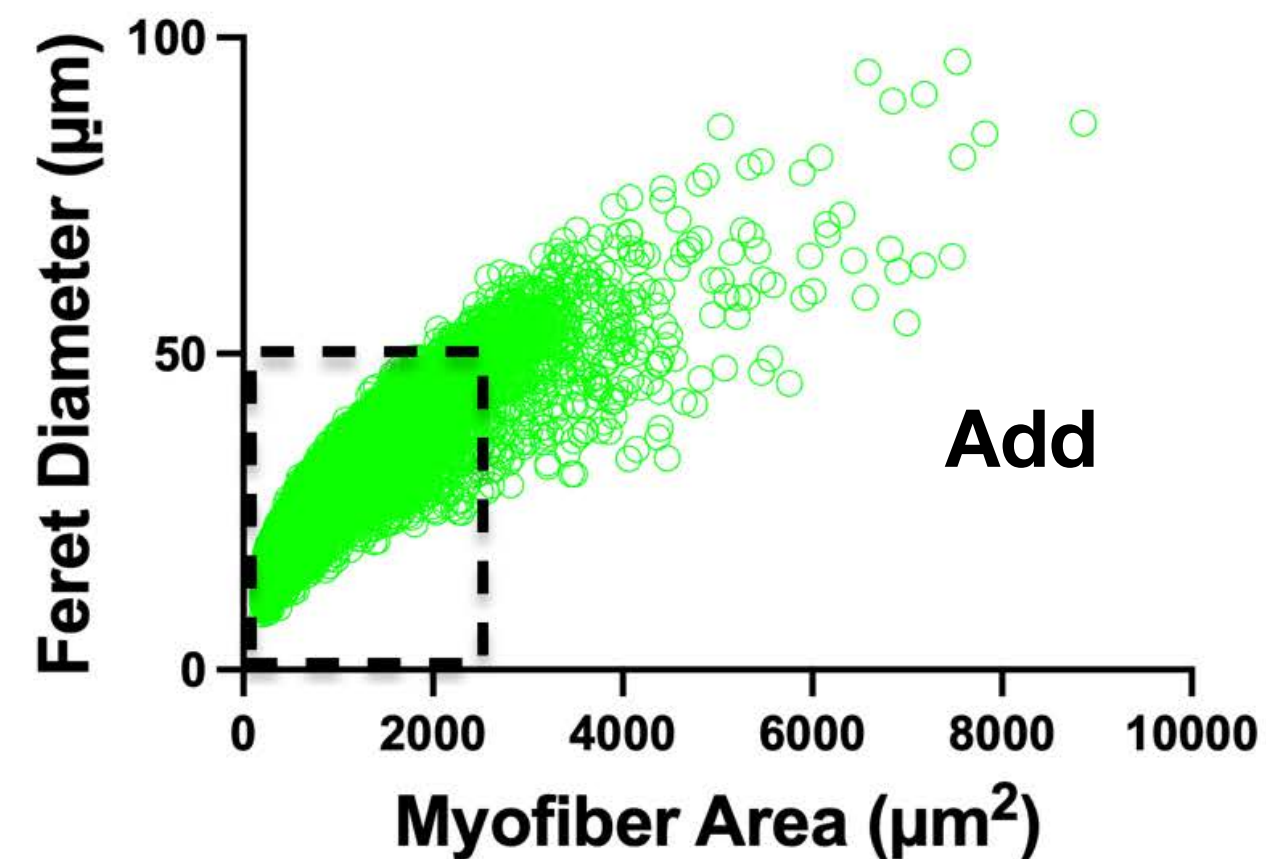
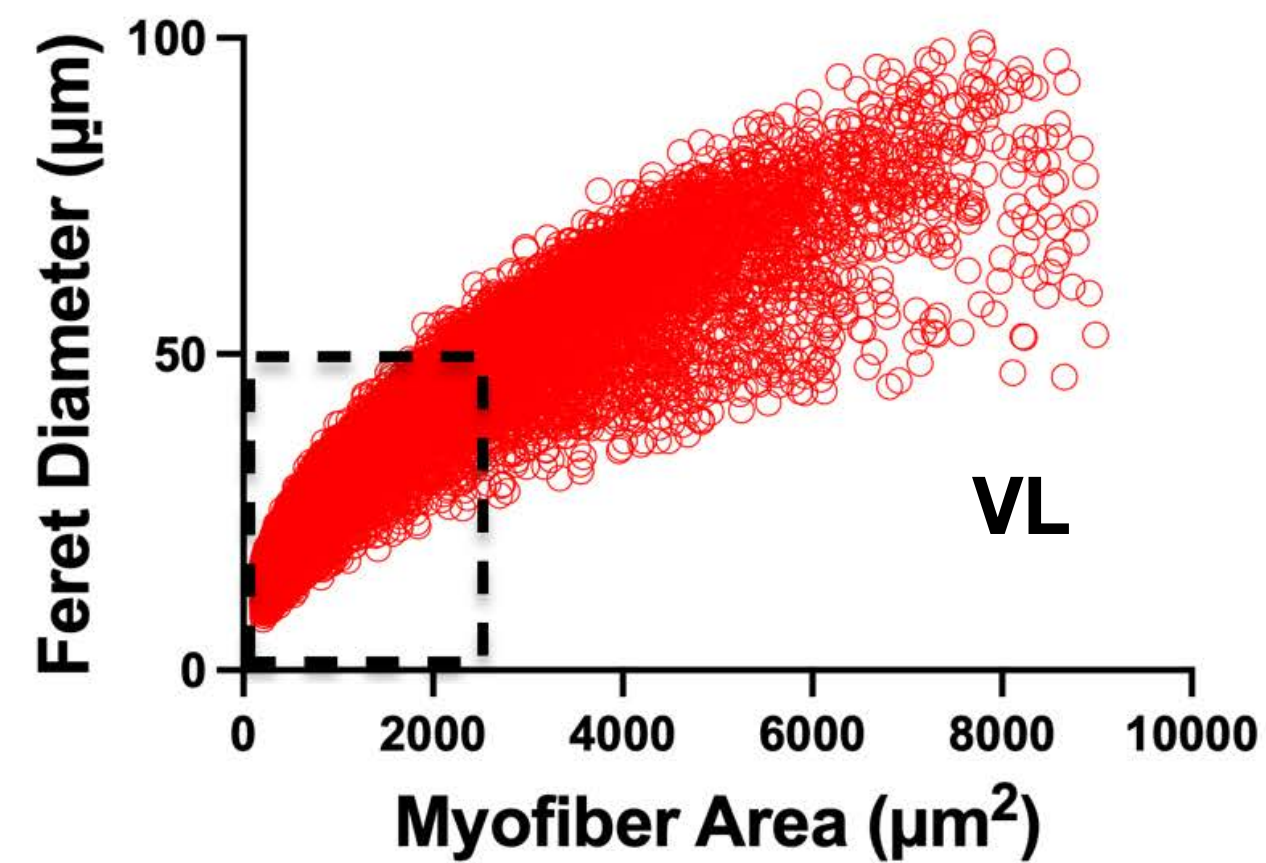
Domenighetti et al 2018



Loomis et al 2025

Edman et al 2026

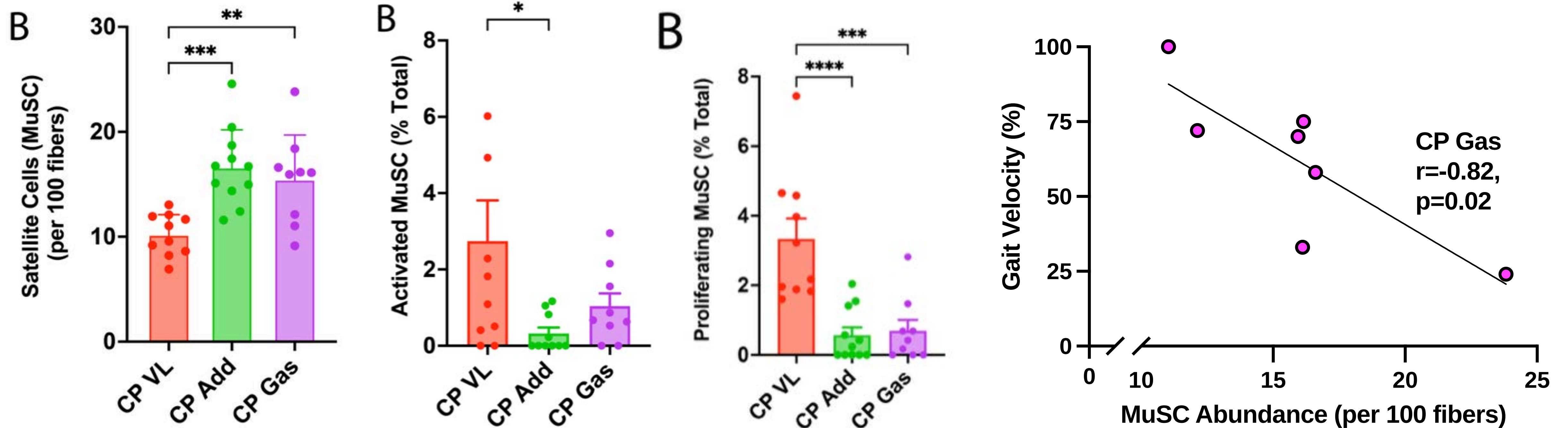
Muscle Stem Cells (MuSC) in situ



Ryan Kahn

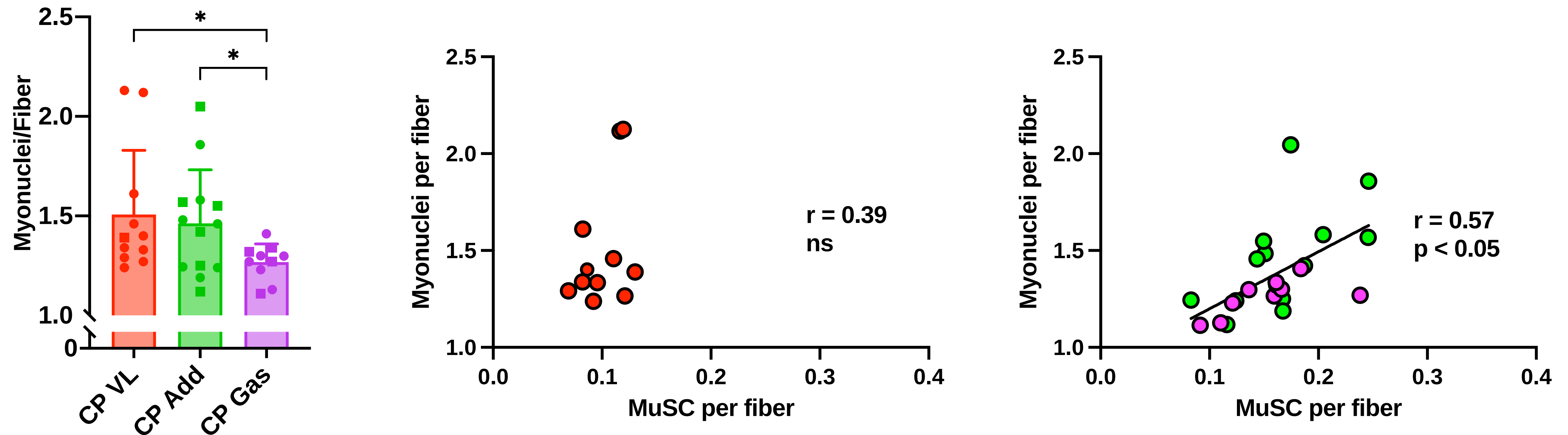
- Muscle characteristics in impaired muscle growth (gastrocnemius, adductor)

Muscle Stem Cells (MuSC)



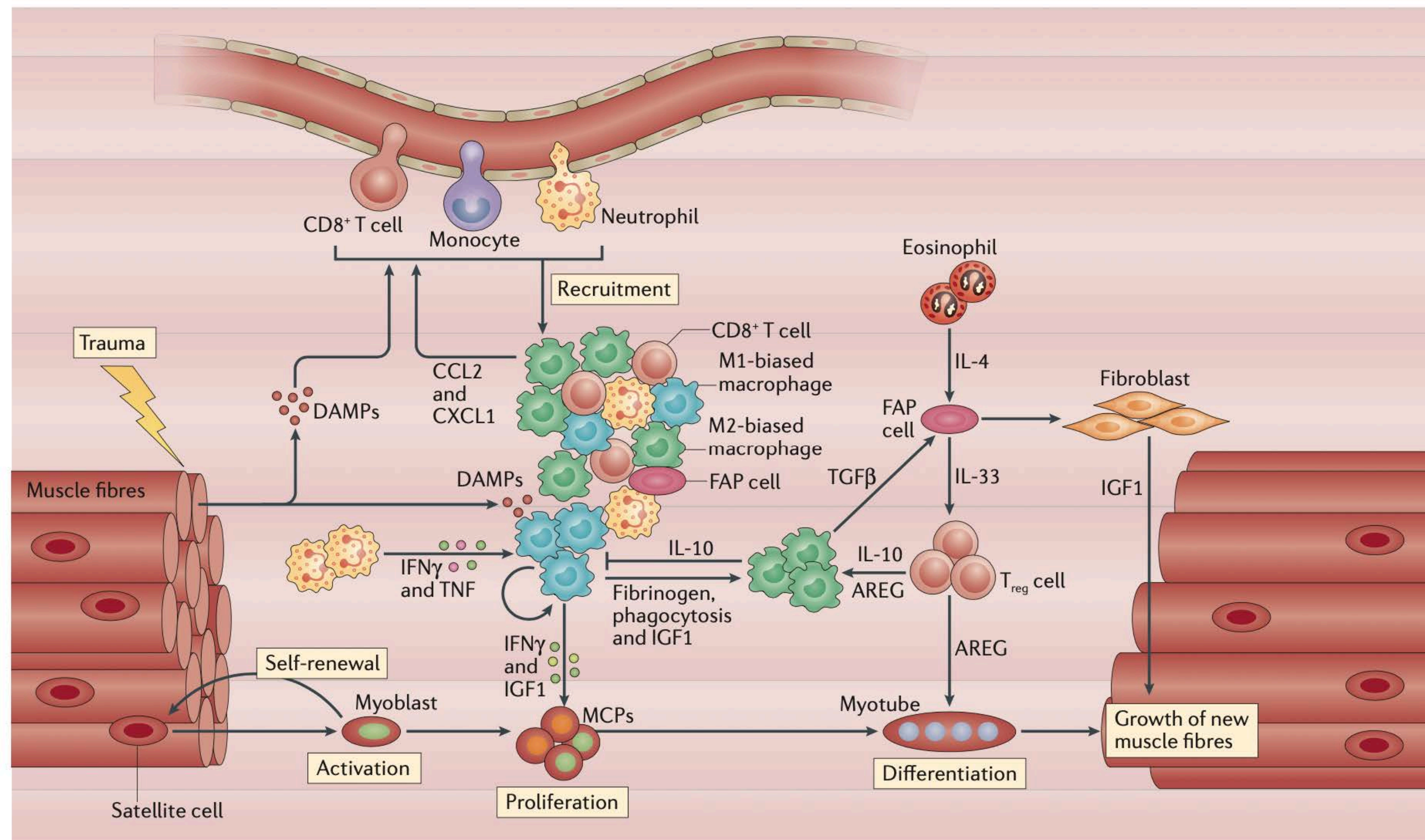
- MuSCs characteristics are different in muscle contractures

MuSC and Myonuclei

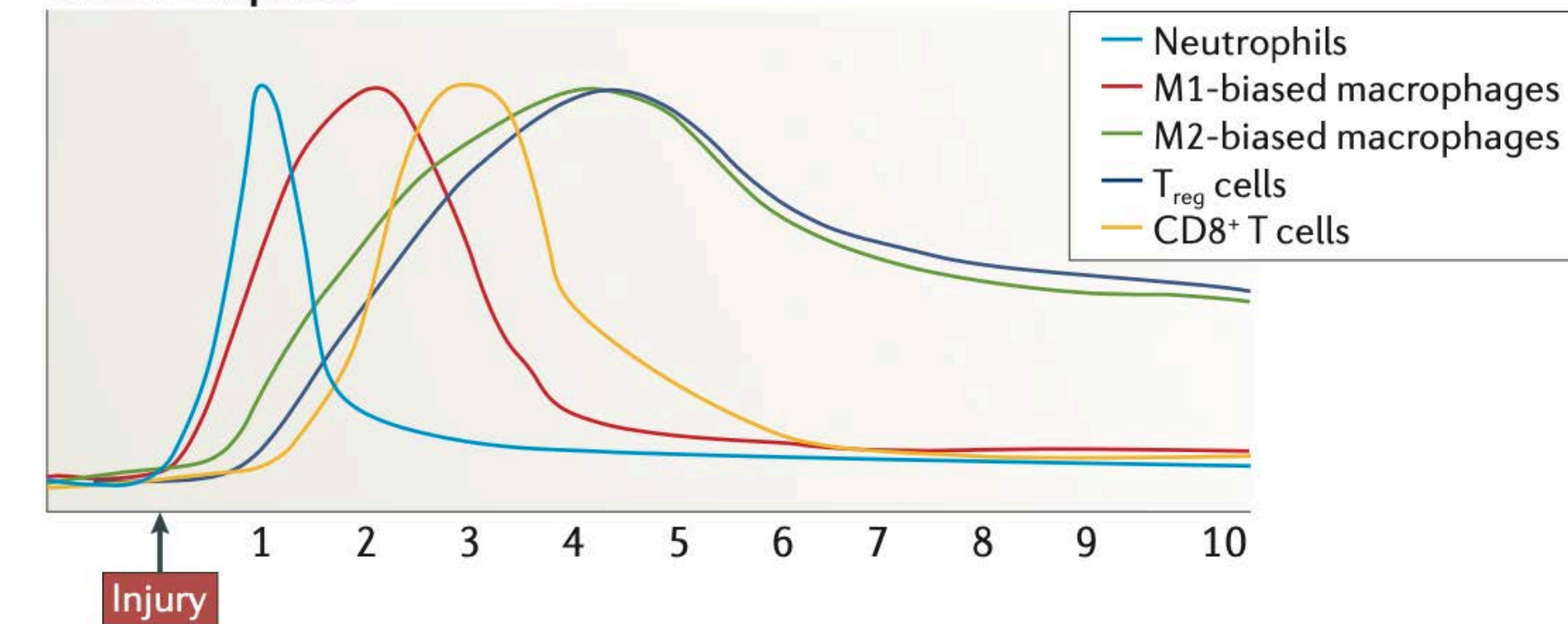


- MuSC abundance across contractured muscles are positively associated with myonuclear number

Muscle Repair

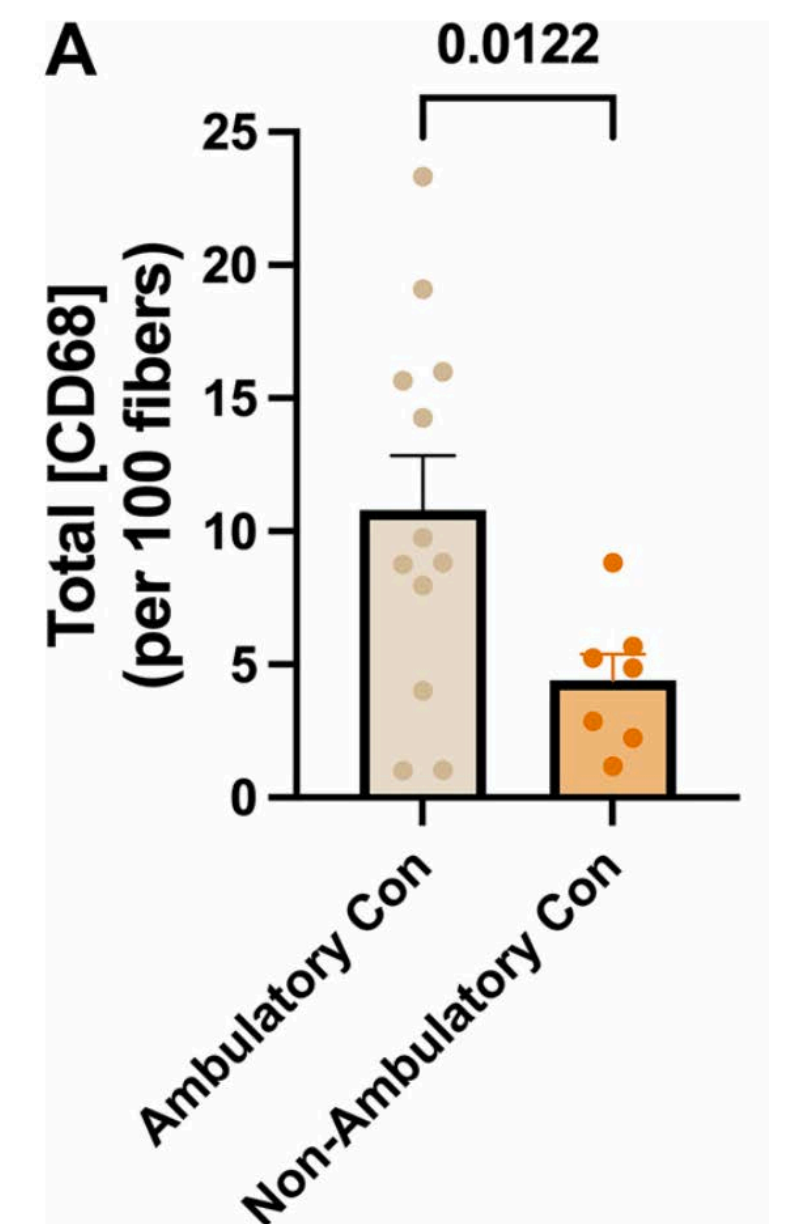
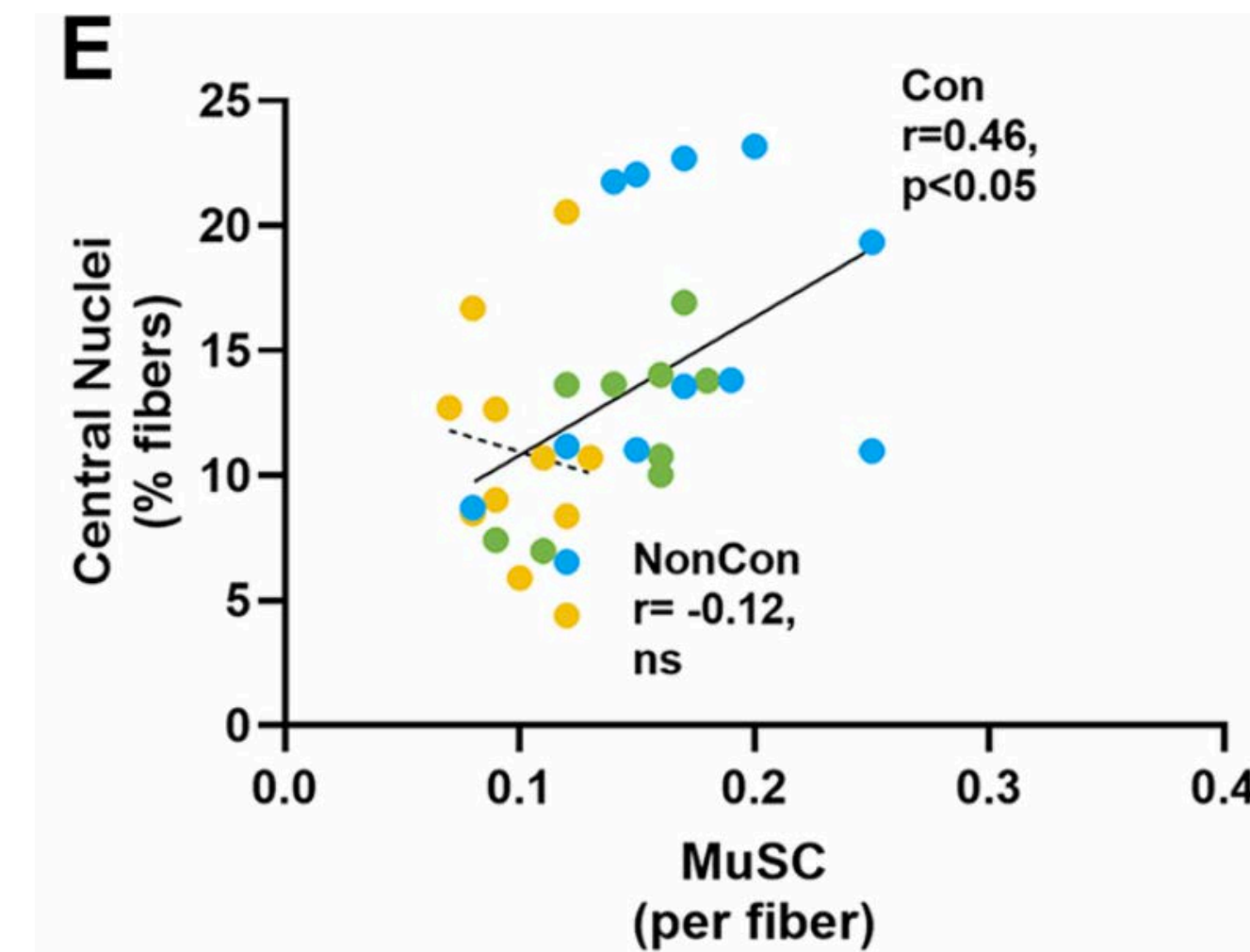
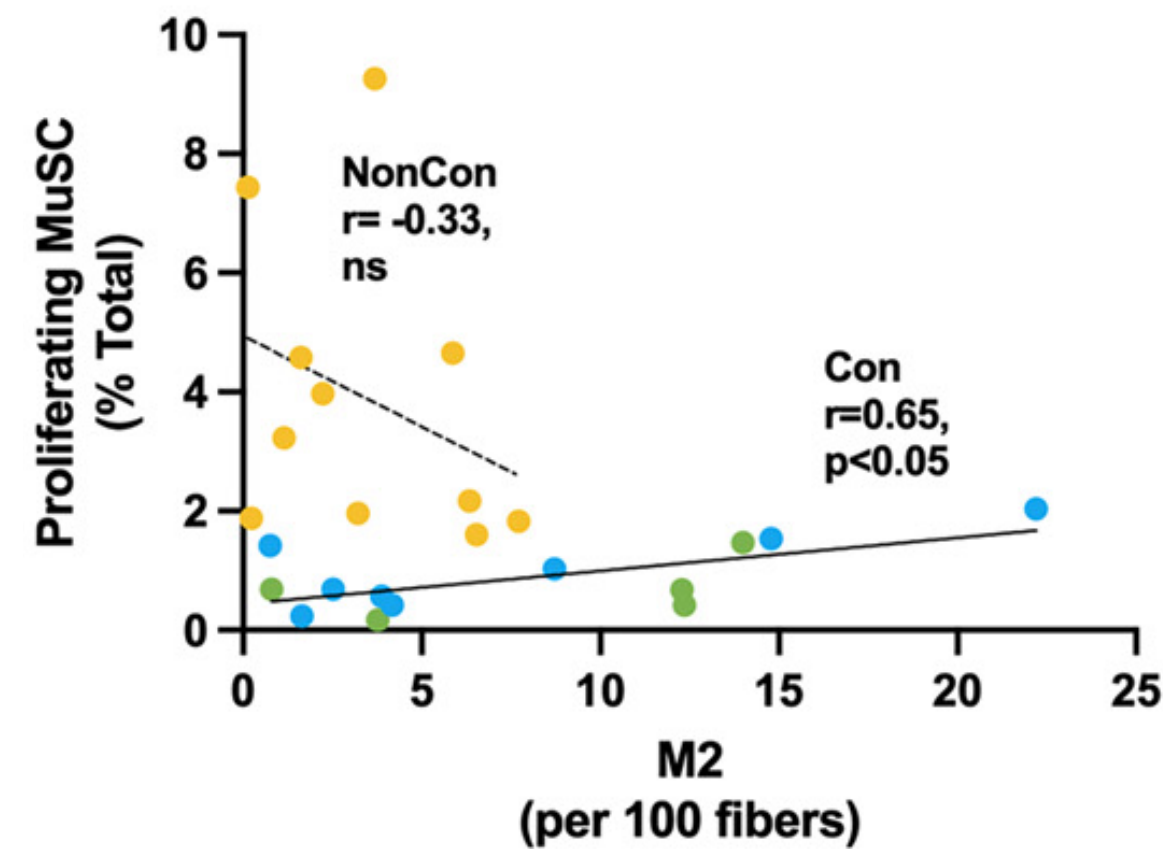
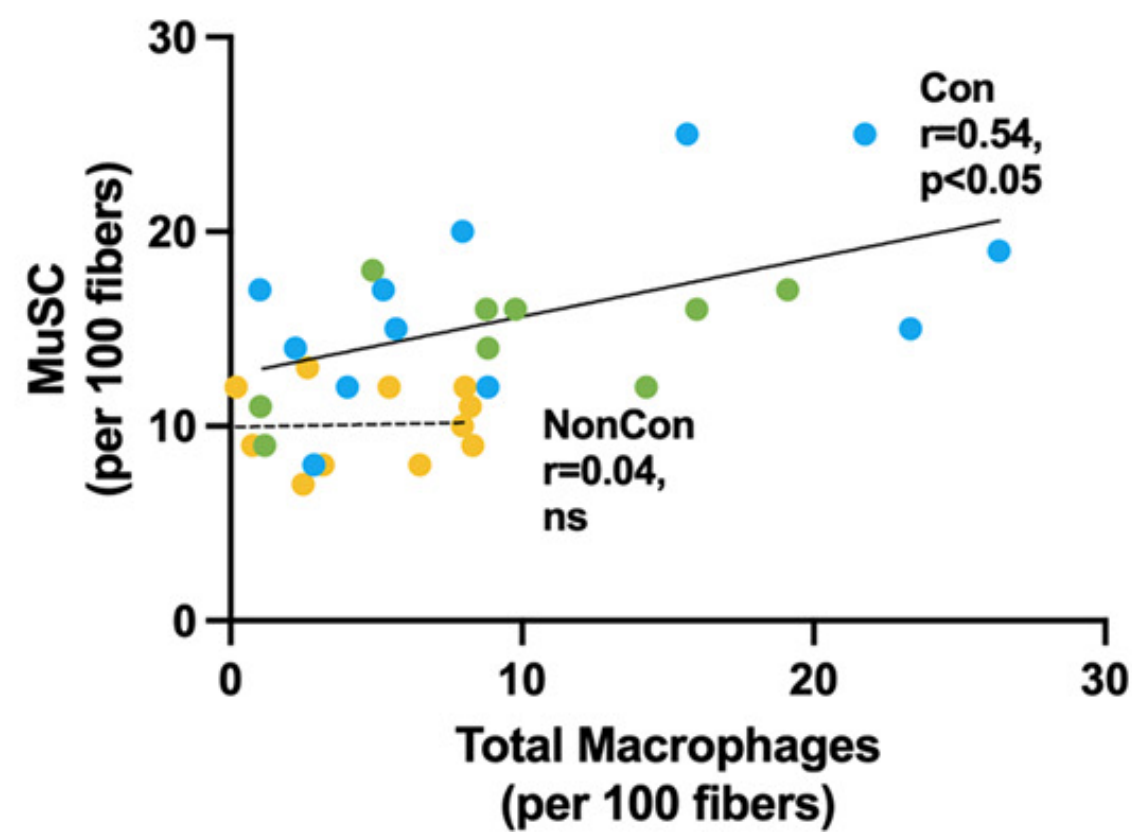


Cellular response



Lupe Meza

Macrophages, MuSC, repair



- Muscle contractures may be showing signs of both repair and growth

Summary

- Lot of unknowns
- Guadalupe Meza
- Ryan Kahn
- Robin Redmond
- Vineeta Swaroop
- Jill Larson
- Neeraj Patel

